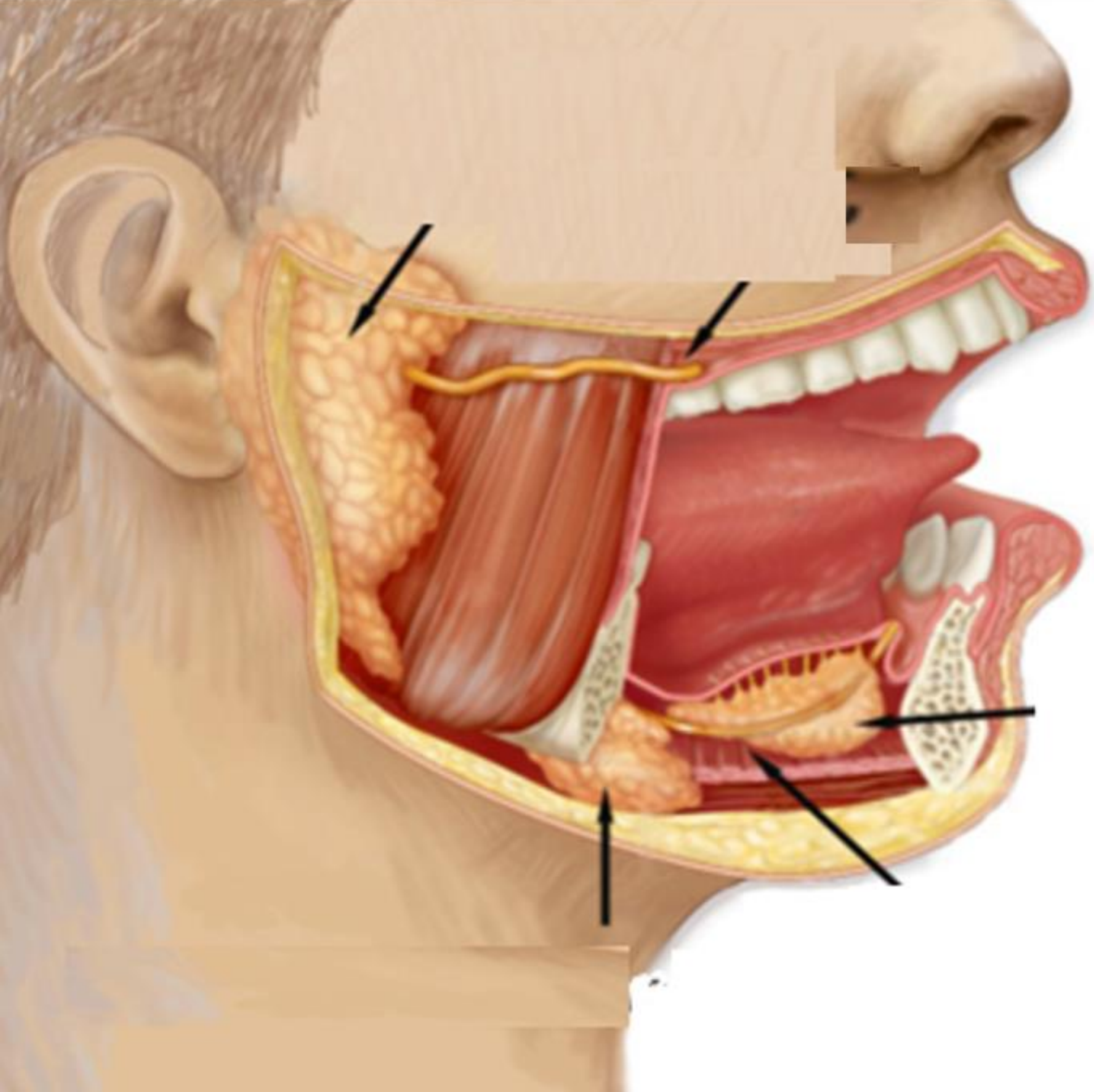


SALIVARY GLANDS (STRUCTURE & DEVELOPMENT)

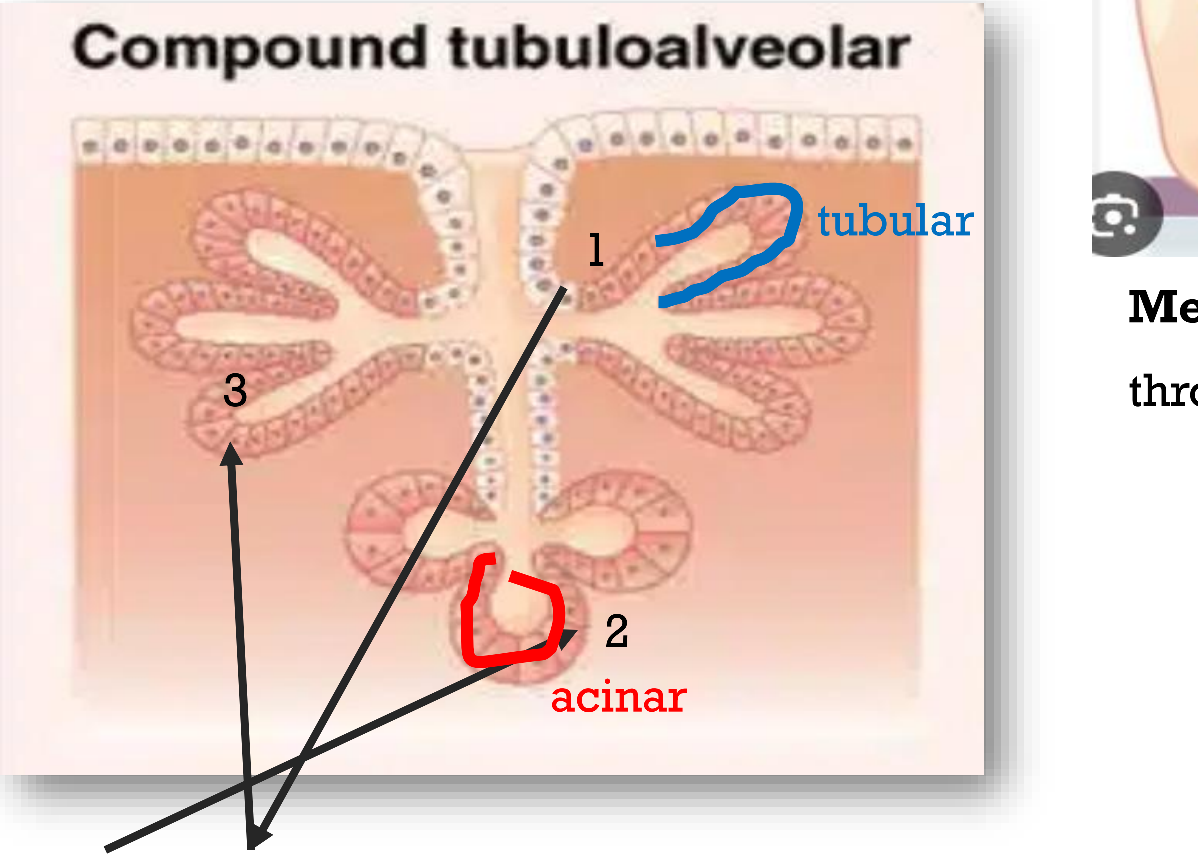
Dr. Misbah Ashraf Moten

Learning Objectives:

- Description of gross anatomy of major & minor salivary glands
- Description of relationships of major & minor salivary glands
- Description of location of major & minor salivary glands
- Understanding of salivary gland development
- Relationship of structure & function of salivary glands to clinical correlations such as xerostomia

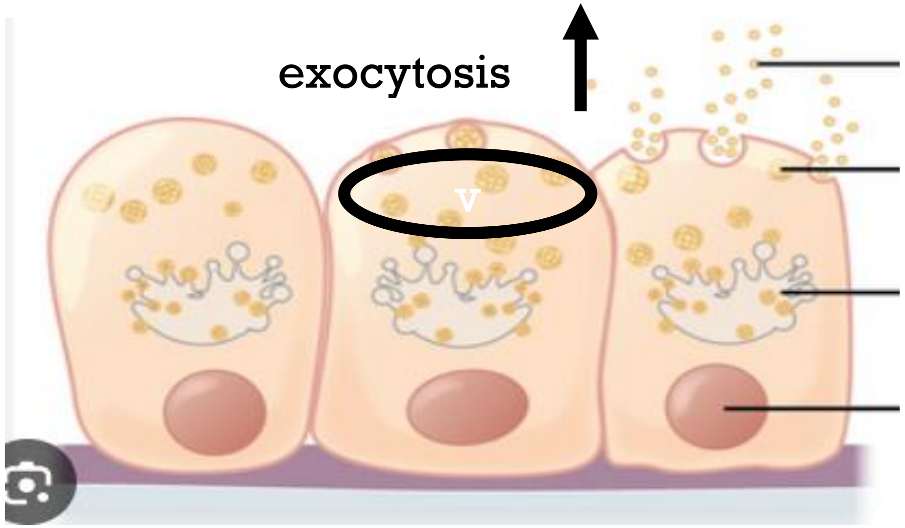
INTRODUCTION:

Salivary Glands are:

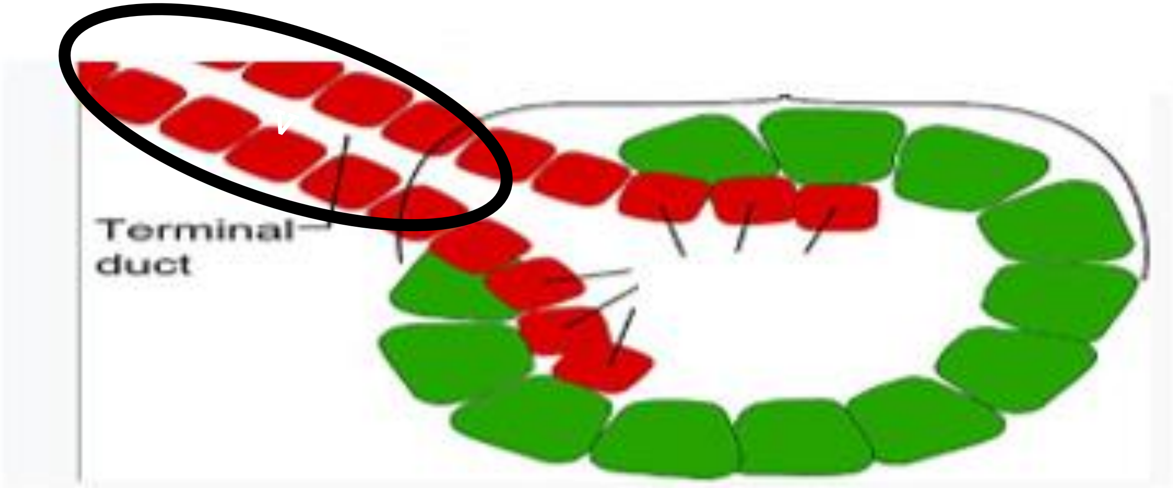


Compound (> 1 tubule entering into main duct)

Tubuloacinar (tubular & alveolar morphology of secreting cells)



Merocrine (secretion comprising of cell only is released through exocytosis of secretory vesicles)

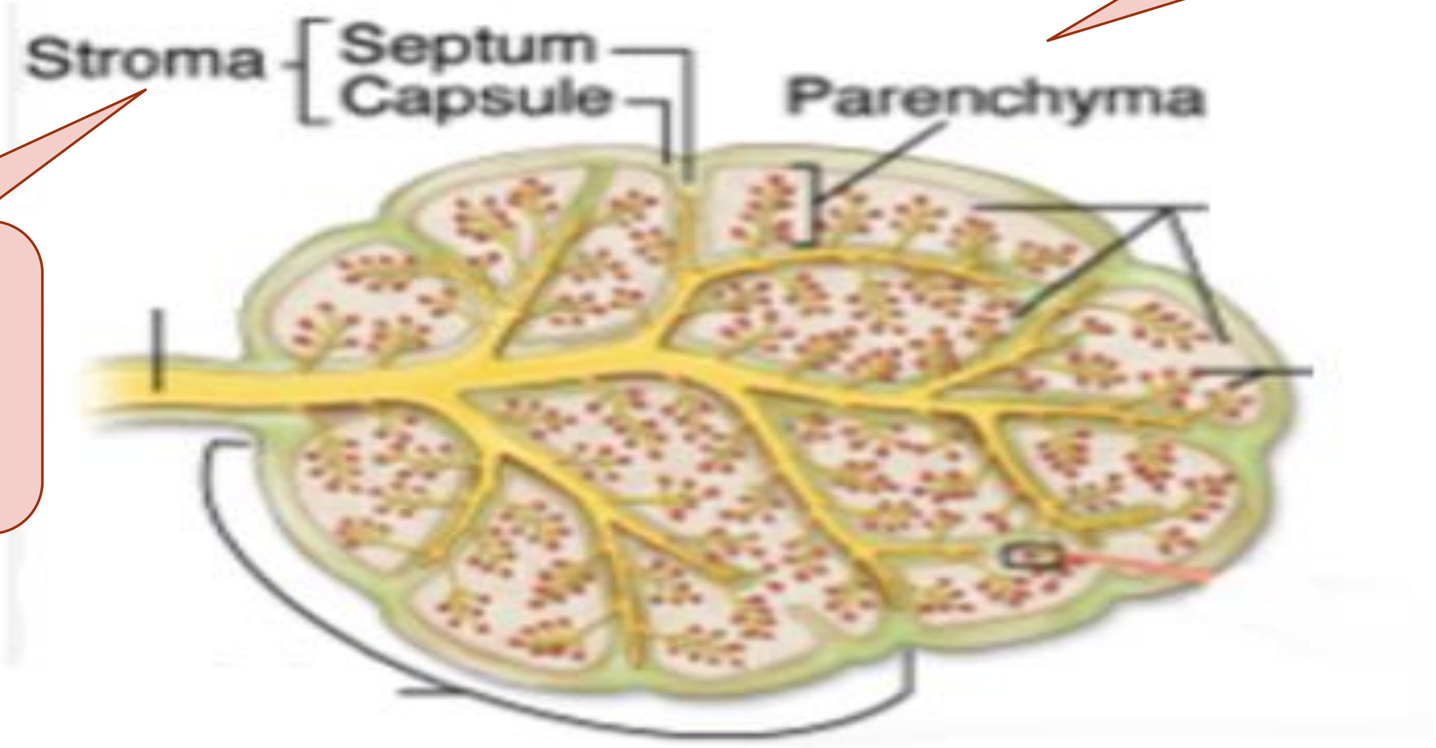


Exocrine (secretion through duct onto free surface such as oral cavity).

ELEMENTS OF SALIVARY GLANDS:

Glandular secretory tissue

Supporting connective tissue



GROSS MORPHOLOGY OF SALIVARY GLANDS:

Connective tissue (stroma based) covering of salivary gland. Function: Protection.

Subdivisions that pass from the capsule, dividing the gland into major lobes.
Lobes are further subdivided into lobules. These carry blood & nerve supply into parenchyma.

Each lobe contains numerous secretory units consisting of clusters of grapelike structures (the acini) positioned around a lumen.
A secretory acinus may be serous, mucous or mixed.

Acini of parenchyma = primary secretion

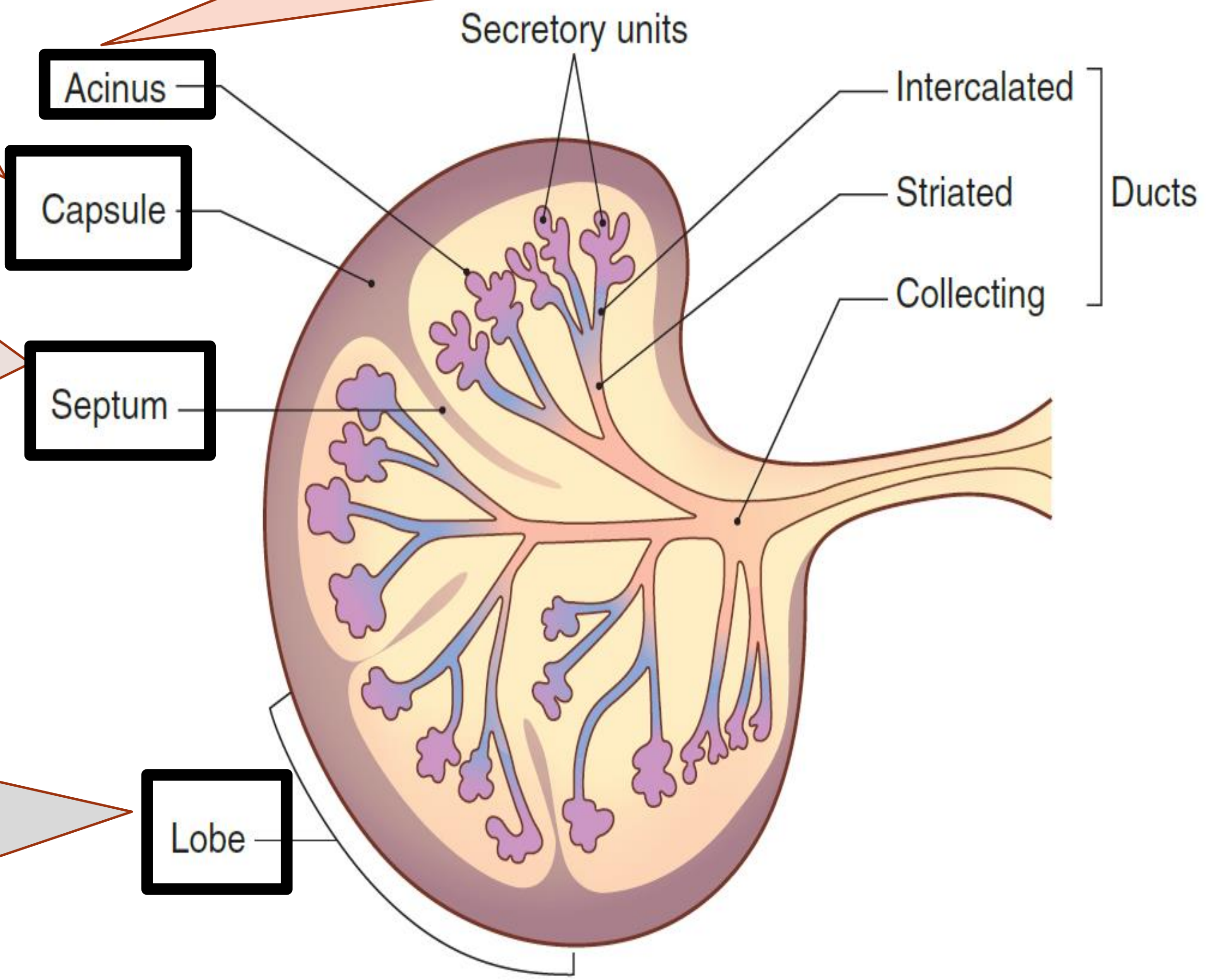


Fig. 16.1 The general organisation of a salivary gland.

GROSS MORPHOLOGY OF SALIVARY GLANDS:

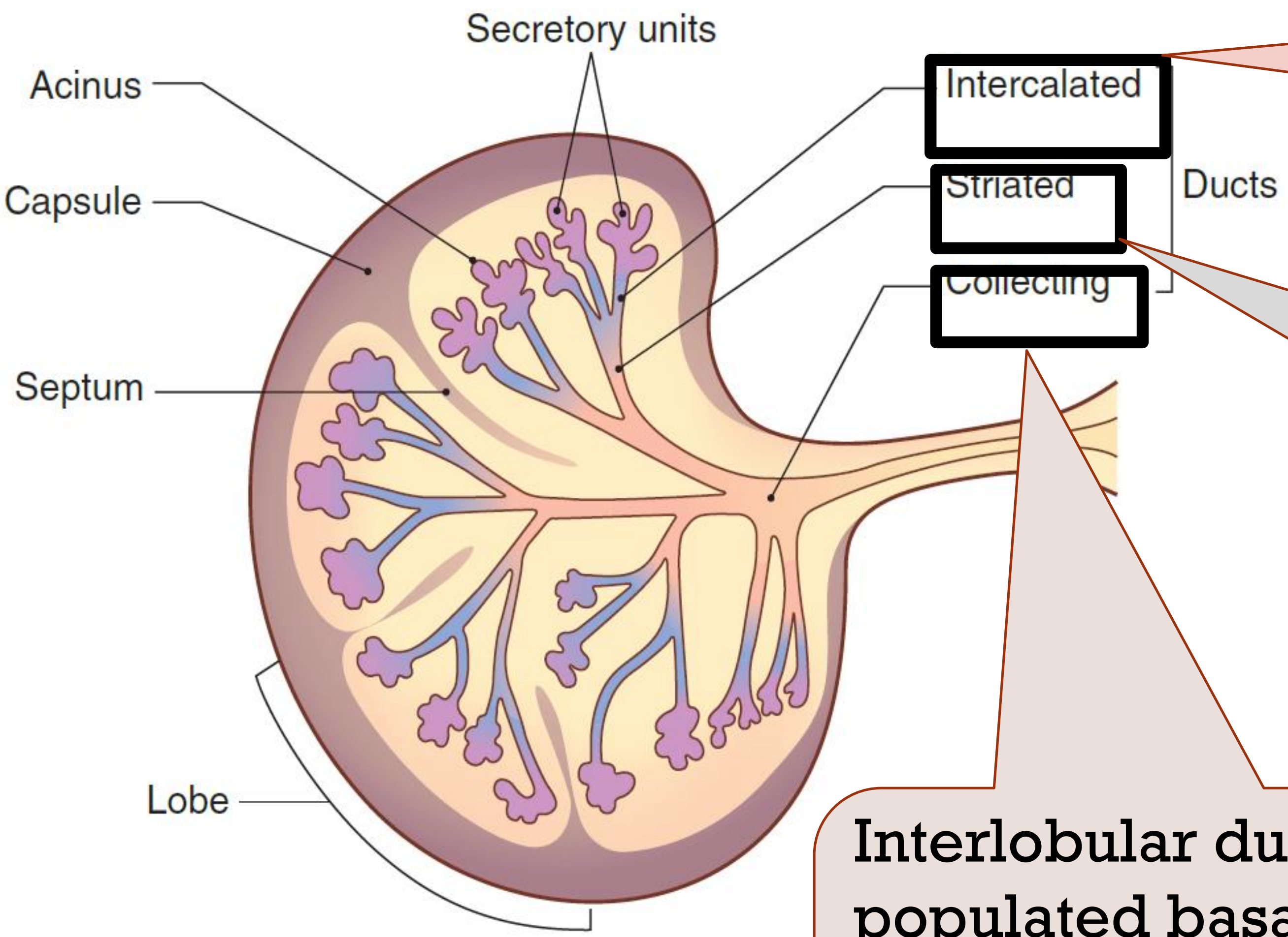
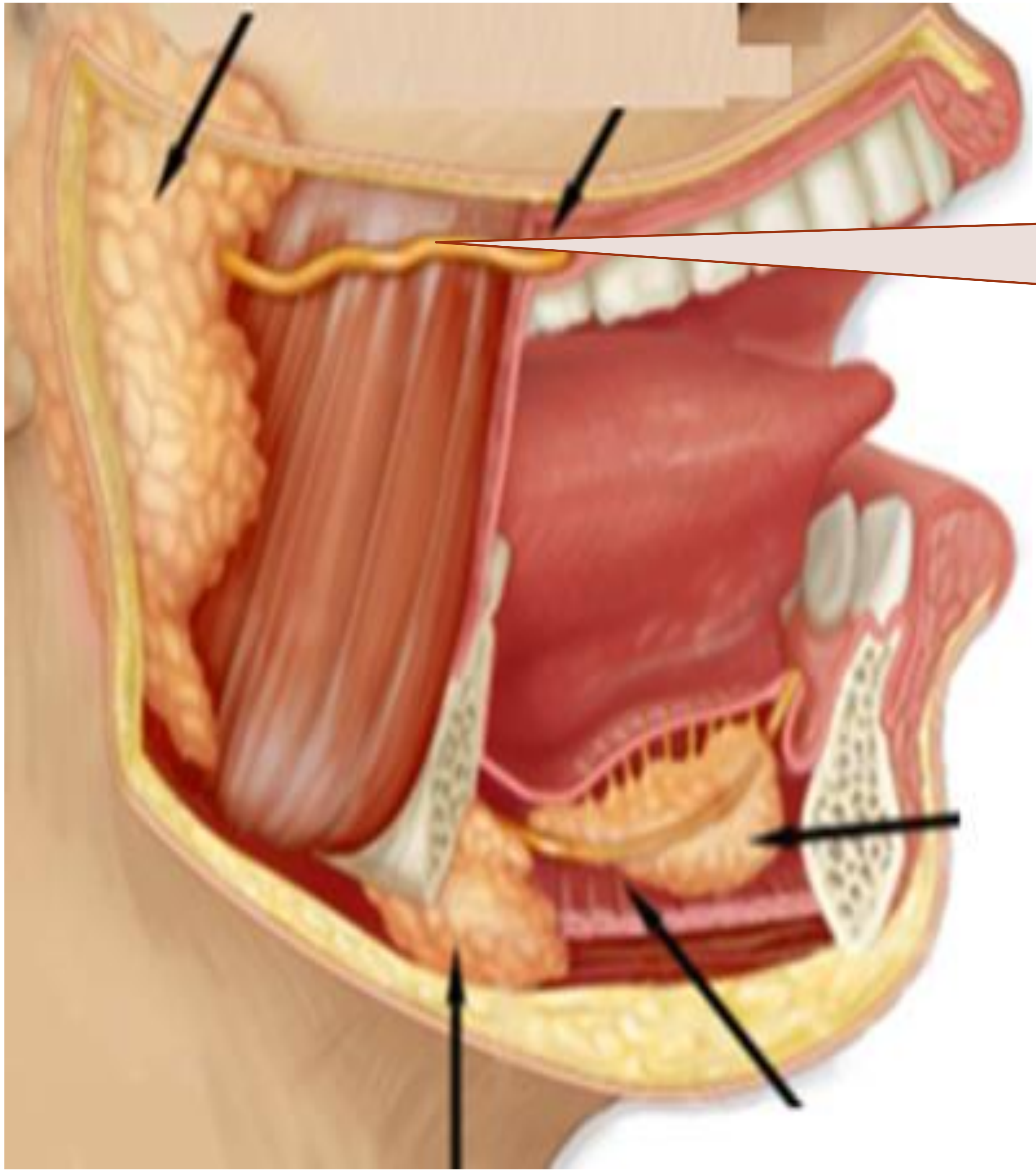


Fig. 16.1 The general organisation of a salivary gland.

Receives secretion from lumen of acinus. It is intralobular

Receives secretion from intercalated duct and is larger in diameter. It is intralobular

Interlobular duct having densely populated basal cells. Receives secretion from striated ducts & joins with other collecting ducts to form main duct.



Collecting ducts join to form main duct at hilum. The main duct carries saliva to mucosal surface and may be lined near its termination by a layer of stratified squamous epithelial cells..

Intercalated duct

CLASSIFICATION OF SALIVARY GLANDS ACCORDING TO SIZE

- **Salivary Glands**
  - **Major Salivary Glands**
    - **Parotid** — protein > glycoprotein; ↓ carbohydrate content
    - **Submandibular**
    - **Sublingual**
    - (Submandibular, Sublingual): Glycoprotein > protein; ↑ carbohydrate content
  - **Minor salivary glands**
    - **Labial**
    - **Lingual**
    - **Palatal**
    - **Buccal**
    - **Glossopalatine**
    - **Retromolar glands**

CLASSIFICATION OF SALIVARY GLANDS ACCORDING TO SECRETION

Salivary Glands

Serous (more protein and less carbohydrate)

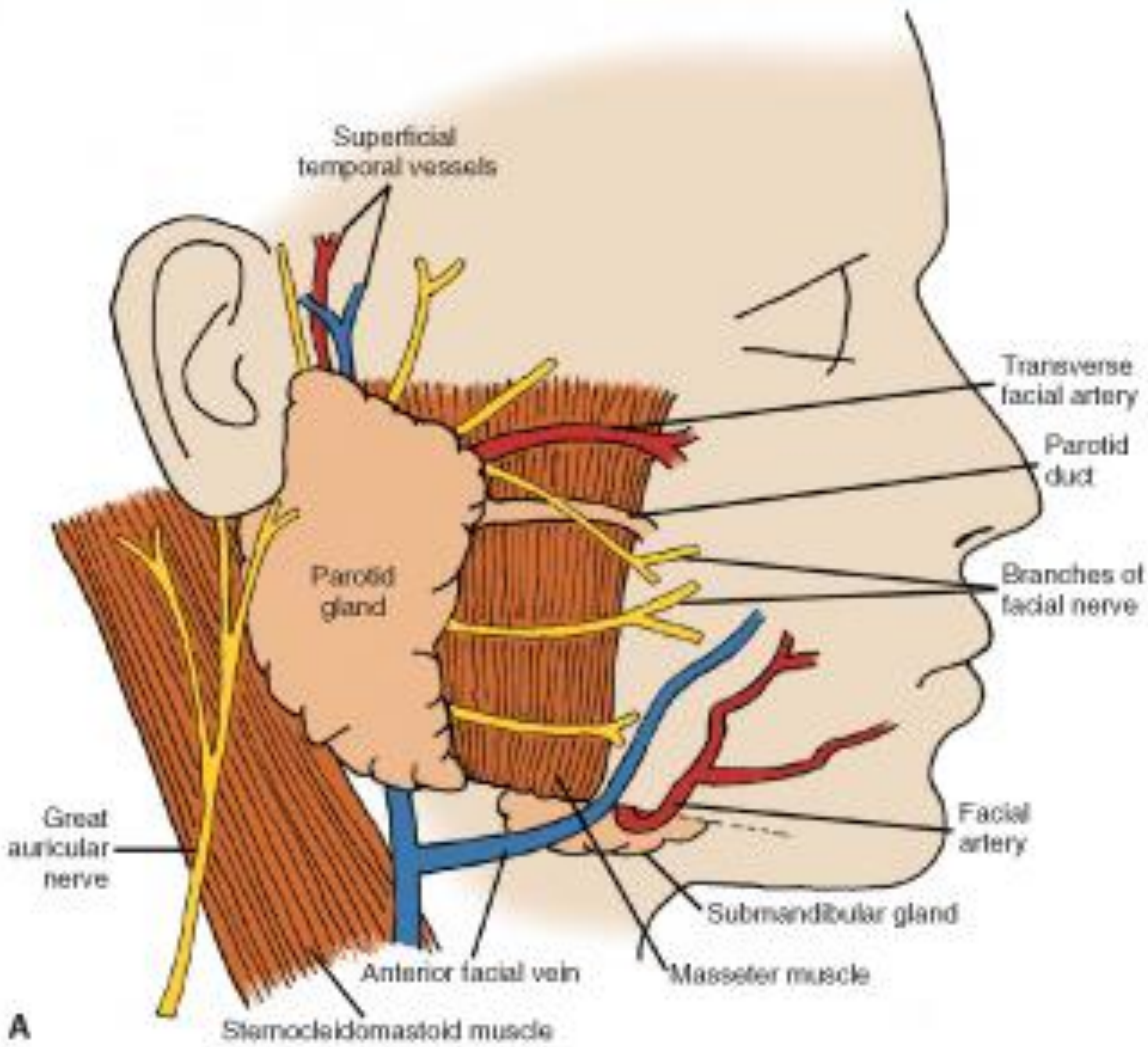
watery saliva rich in enzymes (amylase), proteins (proline-rich proteins) and glycoproteins

Mucous (more protein and less carbohydrate)

contain highly glycosylated substances called mucins

Mixed

Contain both mucous and serous secretions



Superficial temporal vessels
Transverse facial artery
Parotid duct
Branches of facial nerve
Parotid gland
Facial artery
Great auricular nerve
Submandibular gland
Anterior facial vein
Masseter muscle
A
Sternocleidomastoid muscle

SALIVARY SECRETION - KEY POINTS:

- Avg salivary flow rate = **0.3 ml/min**
- Low level secretion = throughout day
- Periodic additions from major glands = meal times.
- Total saliva secreted per day = **500-750ml (90% from major salivary glands)**
- Very small contribution to pooled saliva = gingival fluid & sebaceous glands.
- **Spontaneous saliva secretion = Sublingual & minor salivary glands**
- **Nerve-mediated secretion (bulk) = parotid) submandibular glands**
- Anestehesia = No secretion

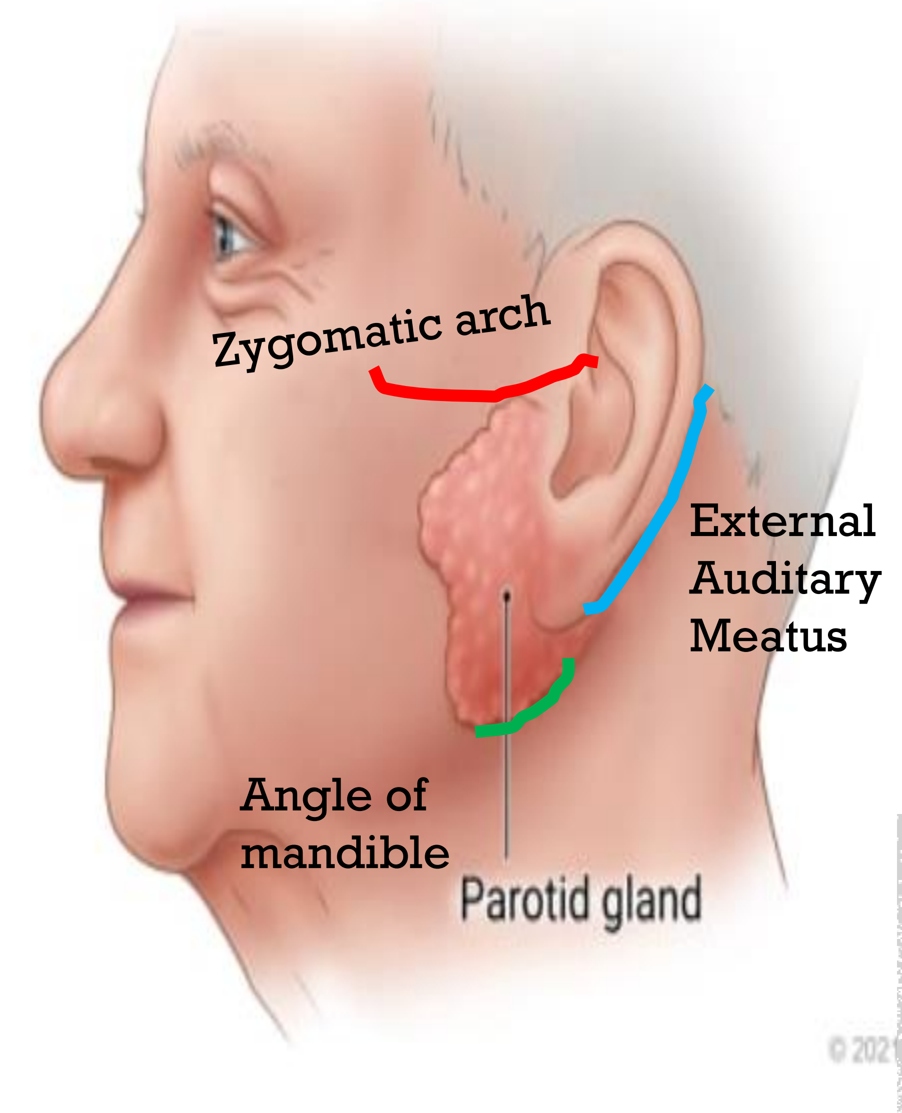
FLOW RATE (ML/MIN)	WHOLE	PAROTID	SUBMANDIBULAR
Resting	0.2-0.4	0.04	0.1
Stimulated	2.0-5.0	1.0-2.0	0.8
pH	6.7-7.4	6.0-7.8	

PAROTID GLAND

Largest salivary gland

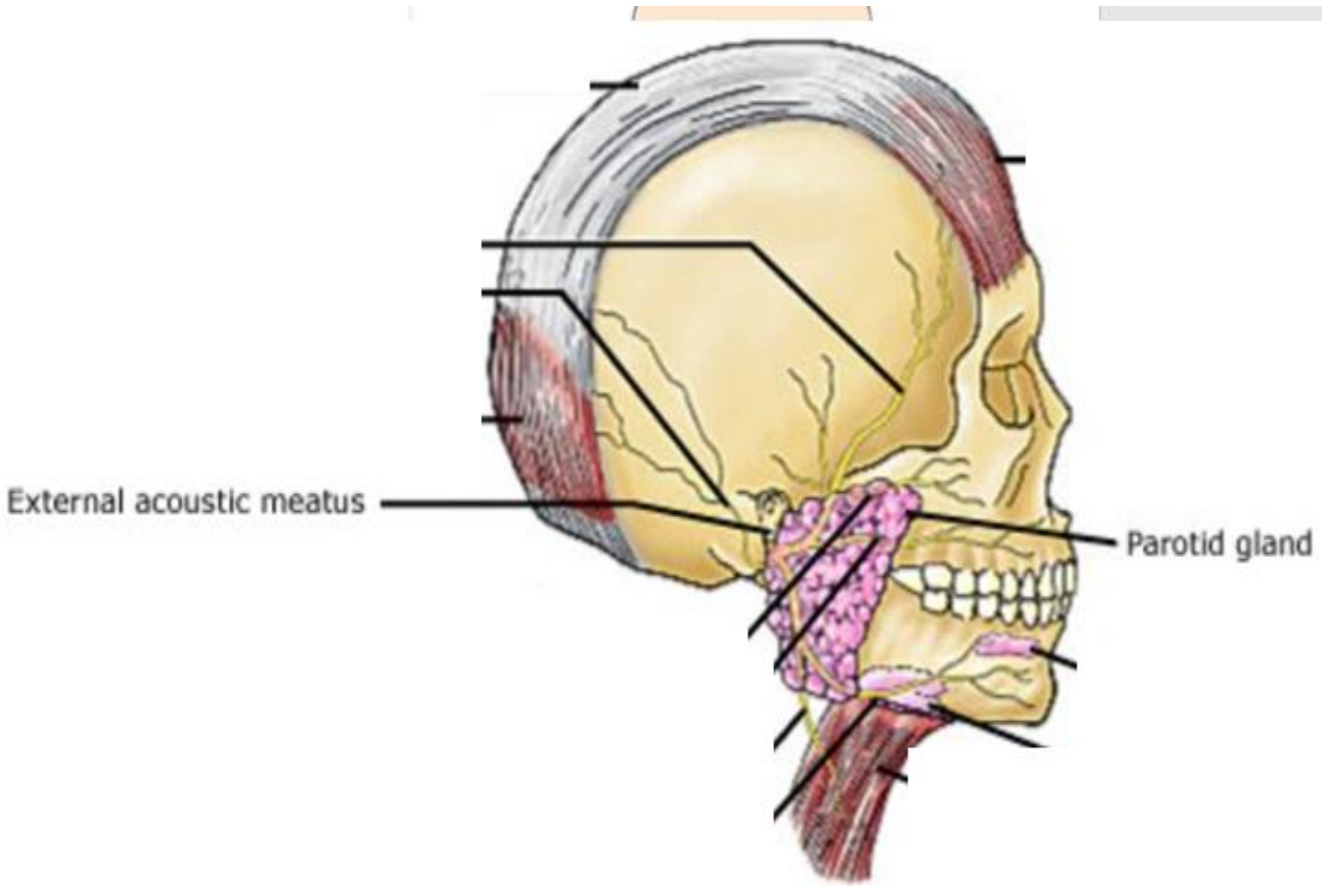
Serous acini- occasionally mucous (infants)

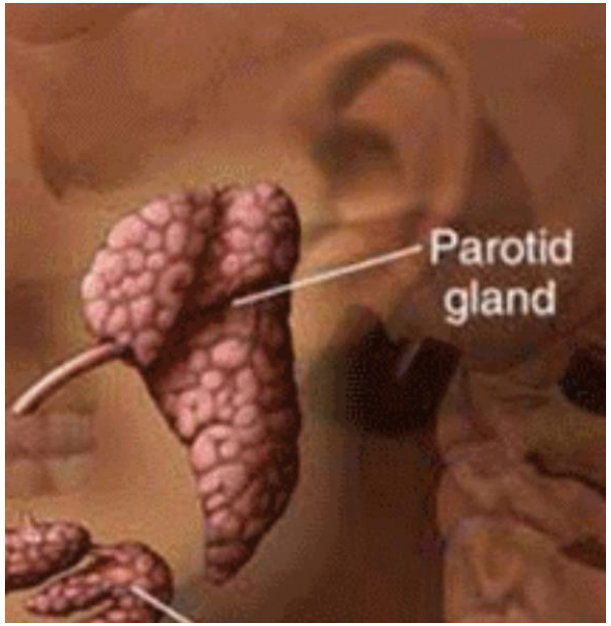
The secretion of this gland is 20-30% of the total salivary volume.



GROSS ANATOMY:

- **Shape:** Pyramidal
- **Capsule**: unyielding tough fibrous capsule (the parotid capsule).
- **Anatmical location:**
- **Anterio-posterior extent:** b/w mandibular ramus & mastoid process.

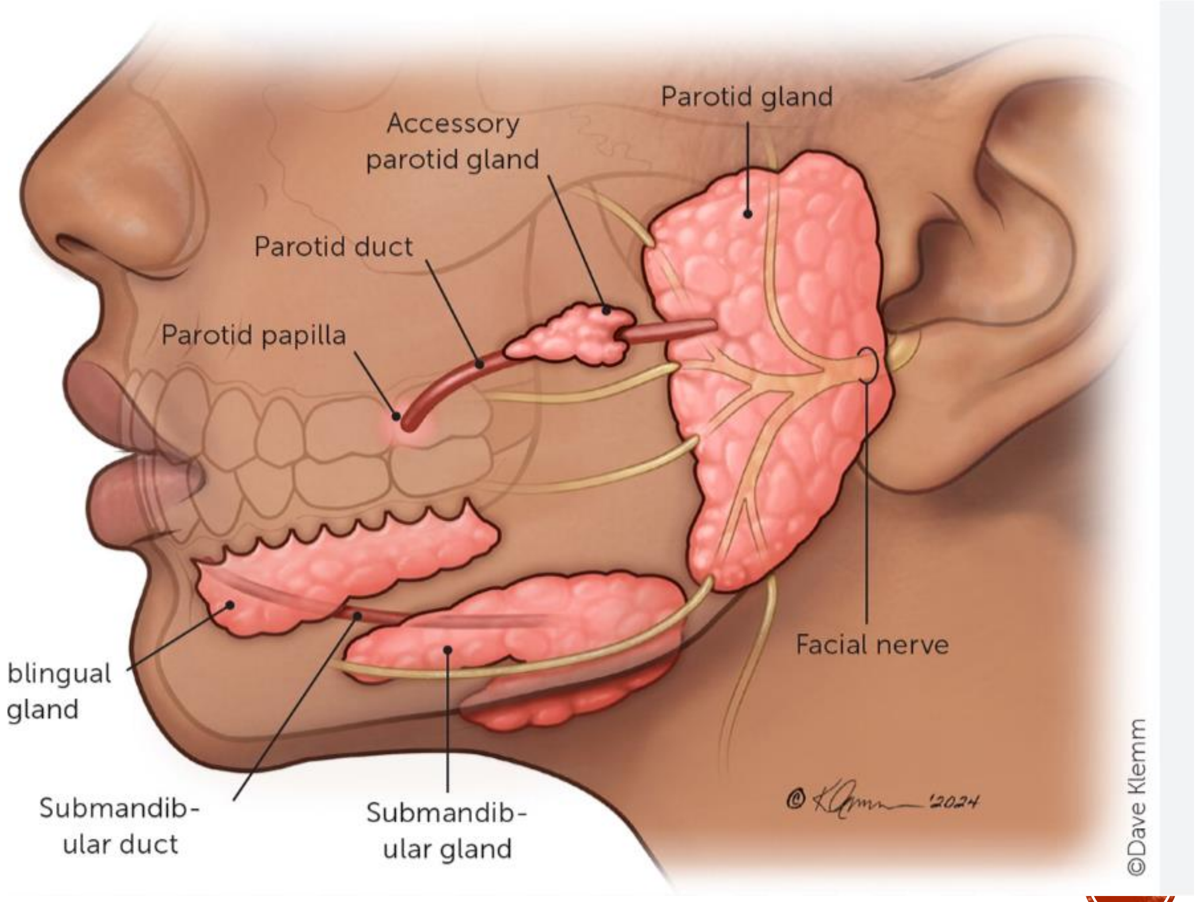




- **Superio-inferior extrent:** apex extends beyond mandibular angle & base closely related to **External Acoustic Meatus**.
- **Deep surface:** rests anteriorly on the ramus & masseter muscle & extends around posterior border of mandible to reach pharynx.

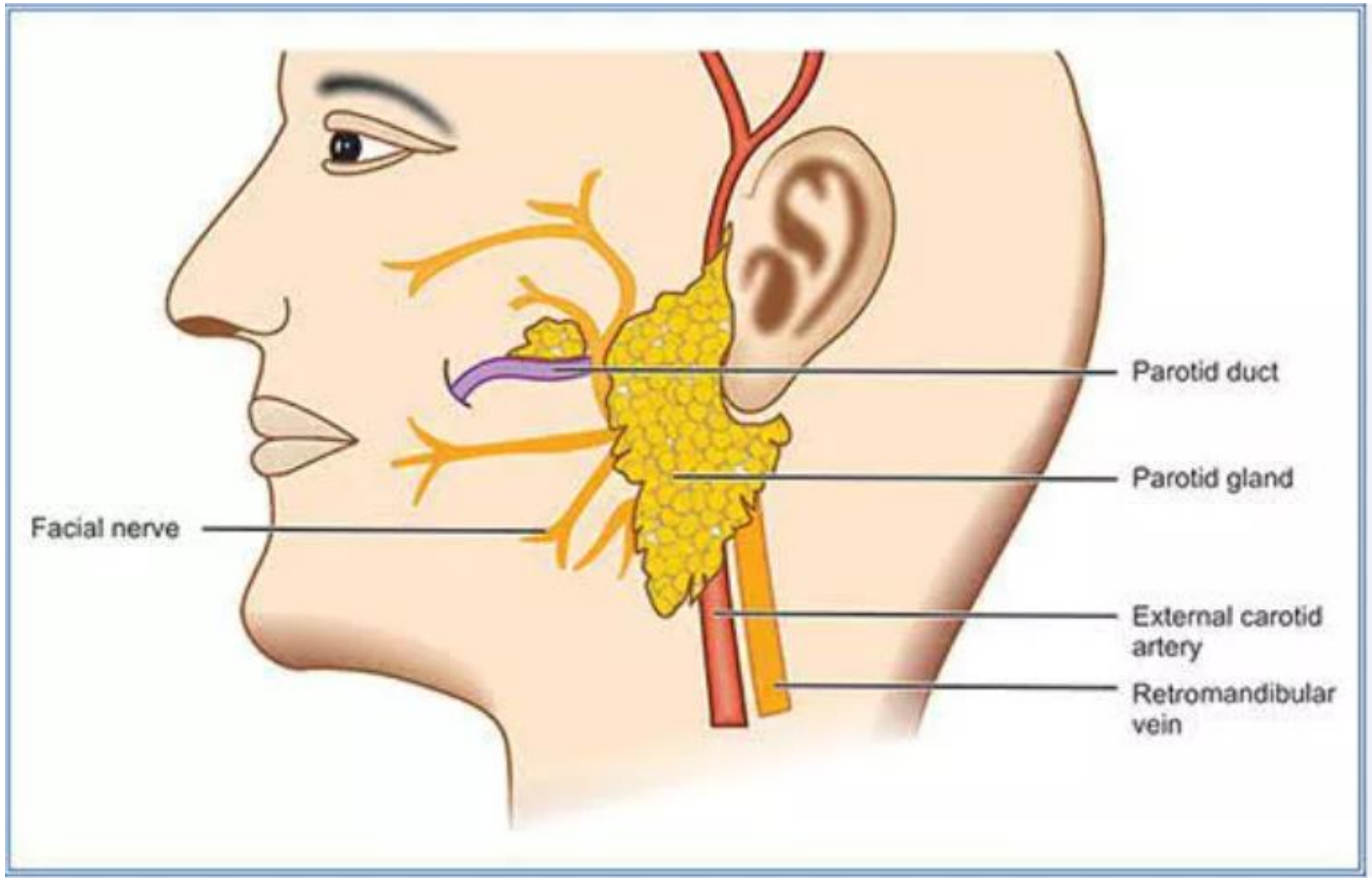
Accessory parotid gland: Lying with the duct on the masseter

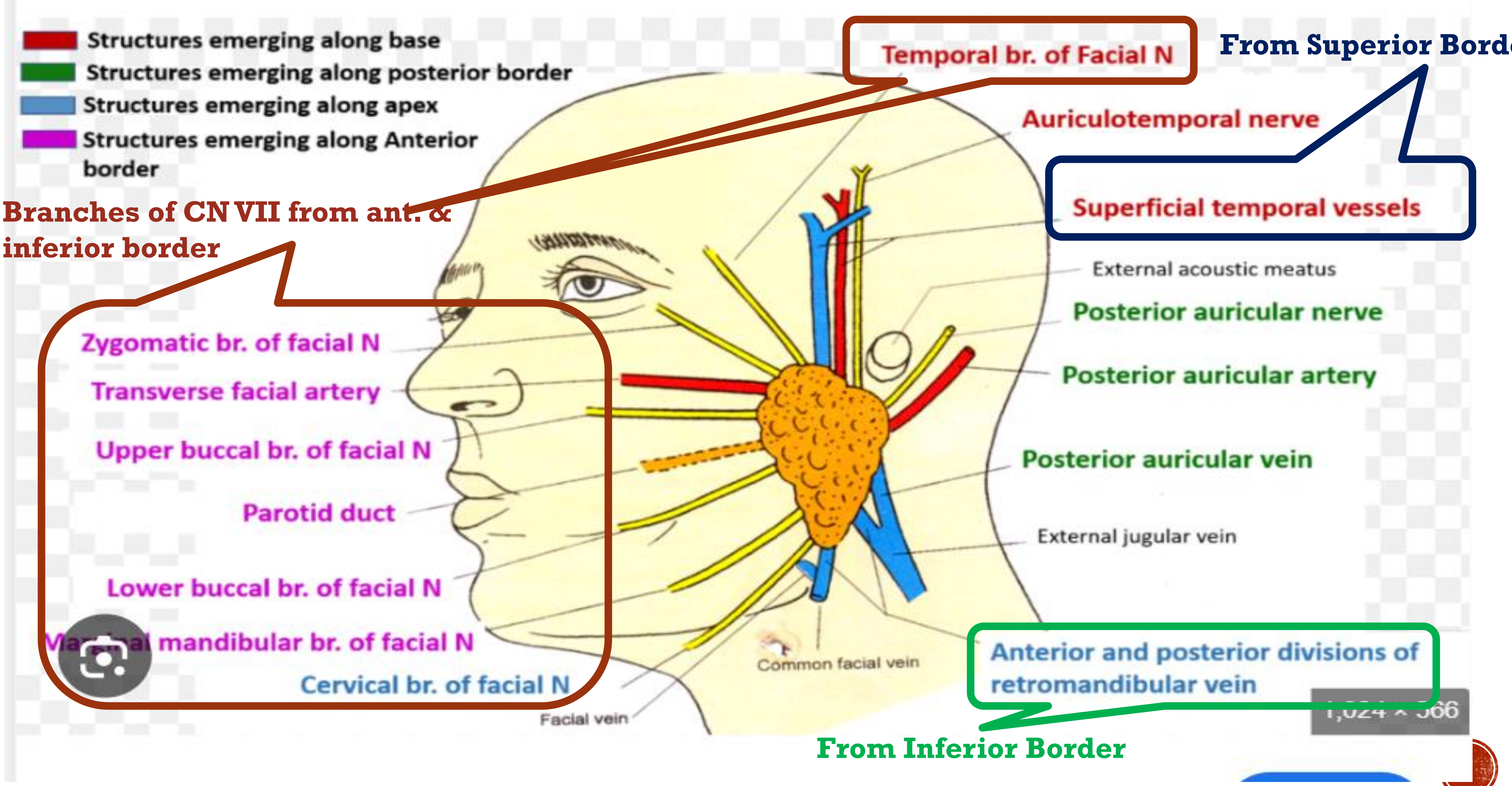
The parotid duct (Stensen's duct): appears at anterior border of gland & passes horizontally across the masseter muscle before piercing the buccinator to terminate in the oral cavity opposite the maxillary second molar.



RELATIONSHIP OF PAROTID GLAND WITH BLOOD VESSELS, NERVES & LYMPH NODES

- Within the gland → External carotid artery, Retromandibular veins, Facial Nerve & Lymph nodes.




Structures emerging along base
Structures emerging along posterior border
Structures emerging along apex
Structures emerging along Anterior border
Temporal br. of Facial N
From Superior Border
Auriculotemporal nerve
Superficial temporal vessels
Branches of CN VII from ant. & inferior border
External acoustic meatus
Zygomatic br. of facial N
Posterior auricular nerve
Transverse facial artery
Posterior auricular artery
Upper buccal br. of facial N
Posterior auricular vein
Parotid duct
External jugular vein
Lower buccal br. of facial N
mandibular br. of facial N
Cervical br. of facial N
Common facial vein
Anterior and posterior divisions of retromandibular vein
Facial vein
From Inferior Border

INNERVATION OF PAROTID GLAND

- ↑ salivary flow during eating → stimulation of sensory receptors, (gustatory, masticatory, psychic, visual olfactory chemoreceptors, thermoreceptive mechanoreceptors & nociceptors)
- Nerve Impulses (salivary secretion control)

  Afferent Arm → Medulla Oblongata

  Efferent Arms → (Sympathetic & Parasympathetic nerves)

Gustatory-Salivatory Reflex

- Stimulation of gustatory receptors
- Predominant in taste buds
- By all types of tastes (salt, sour, sweet, bitter & umami)

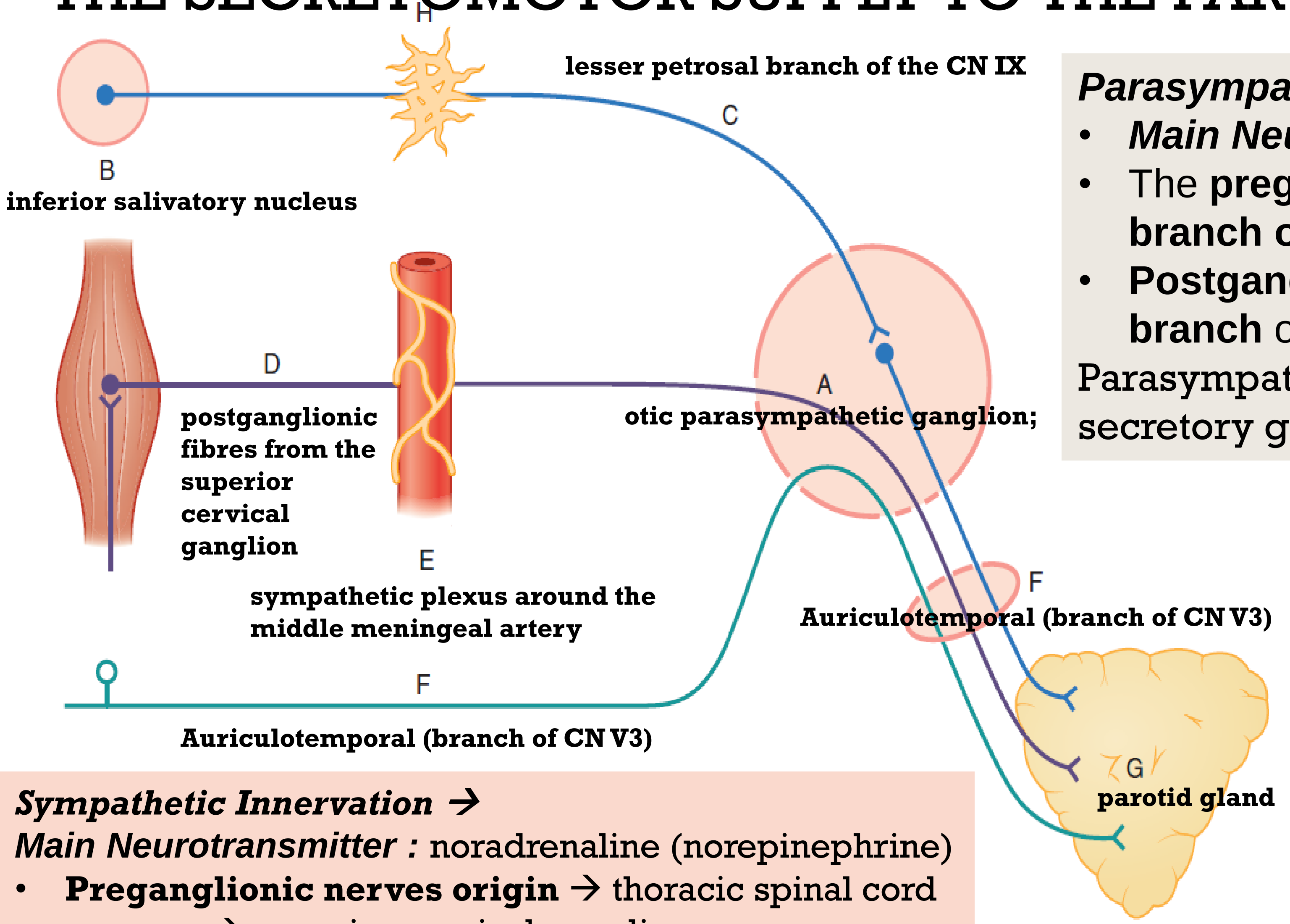
Masticatory-Salivtory Reflex

- Stimulated by intraoral mechanoreceptors (esp. PDL mechanoreceptors)
- Chewiung on one side → ↑ salivary flow on that side as compared to nonchewing side
- Saliva produce directly proportional to masticatory forces

Olfactory-Salivary Reflex

- Submandibular/sublingual gland (not parotid)

THE SECRETOMOTOR SUPPLY TO THE PAROTID SALIVARY GLAND.



Parasympathetic innervation -→

- ***Main Neurotransmitter :*** acetylcholine
- The **preganglionic fibres** from **lesser petrosal branch of CN IX** synapse in **otic ganglion** &
- **Postganglionic fibres** travel into **auriculotemporal branch** of **CN V3**

Parasympathetic drive causes formation & secretion of secretory granules & fluid by the secretory units;

Sensory Innervation (Parotid Capsule) →

- **great auricular nerve**, a branch of the **cervical plexus** (**roots of cervical plexus → CN II & III anterior primary rami**).

Sympathetic Innervation →

Main Neurotransmitter : noradrenaline (norepinephrine)

- **Preganglionic nerves origin** → thoracic spinal cord
- synapse → superior cervical ganglion
- postganglionic neurons → travel with arterial blood supply
- sympathetic drive usually increases the output of preformed components from the cells.

NEUROTRANSMITTORS & NEUROPEPTIDES :

NEUROTRANSMITTORS (endogenous chemical messengers that allow neurons to communicate with each other and with target effector cells - muscles, glands, across a small gap called a synapse)

- ***Parasympathetic innervation → Main Neurotransmitter : Acetylcholine***
- ***Sympathetic Innervation → Main Neurotransmitter : noradrenaline (norepinephrine)***
- Embryologically, the transmitters are likely to influence the genetic expression of the glandular cells
- ***In electron micrographs, the conventional neurotransmitters are contained in small vesicles***

OTHER NEUROPEPTIDES

- Each axon contains arrays of neuropeptides, such as: (Vasoactive intestinal polypeptide, substance P & calcitonin-gene-related peptide)
- These neuropeptides are not necessarily uniform for each type of nerve, nor are all present within every nerve of the same type.
- The cell types are likely to influence the neuropeptides in the axons innervating them.
- ***In electron micrographs, neuropeptides are contained in larger, dense-cored vesicles***

NEUROEFFECTOR SITES:

- Neuro effectors are the target cells that respond to neurotransmitter signals, resulting in a physiological change.
- ***Hypolemmal:*** Some neuroeffector sites occur beneath the basal lamina in direct contact with the plasma membrane
- ***Epilemmal:*** Neuroeffector sites that remain outside the basal lamina –epilemmal

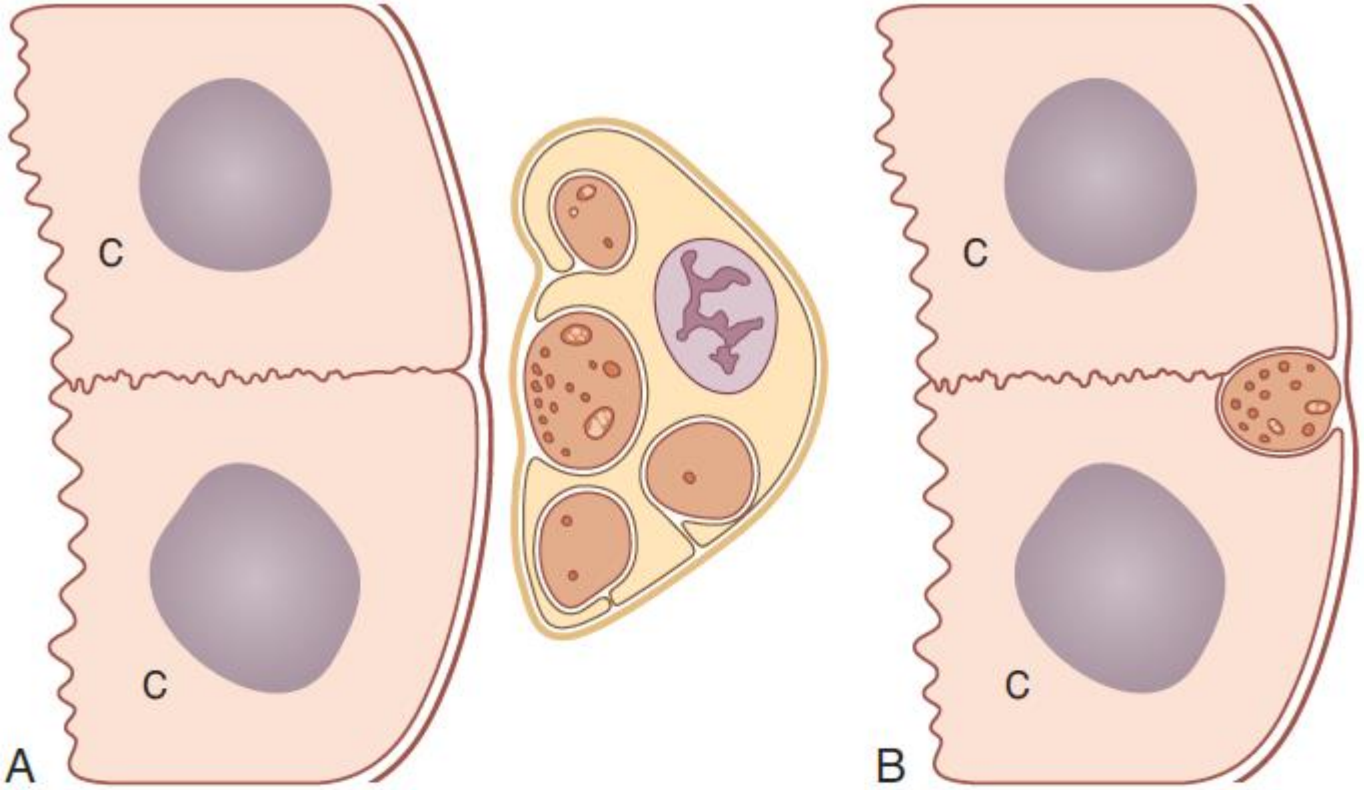
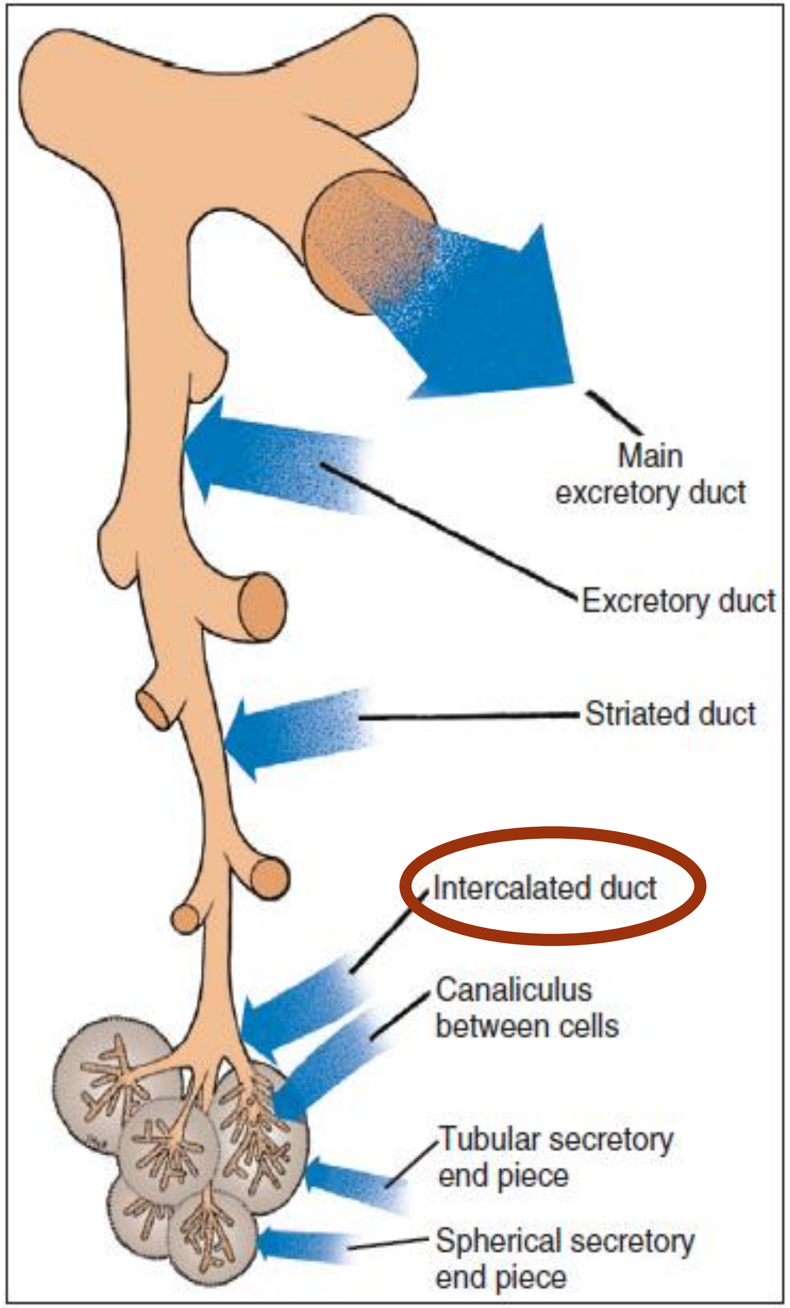


Fig. 16.12 Neuroeffector relationships in salivary glands. A = epilemmal; B = hypolemmal; C = parenchymal cells. Courtesy Prof. J R Garrett and Karger Press.

DUCTAL SYSTEM

INTERCALATED DUCTS:

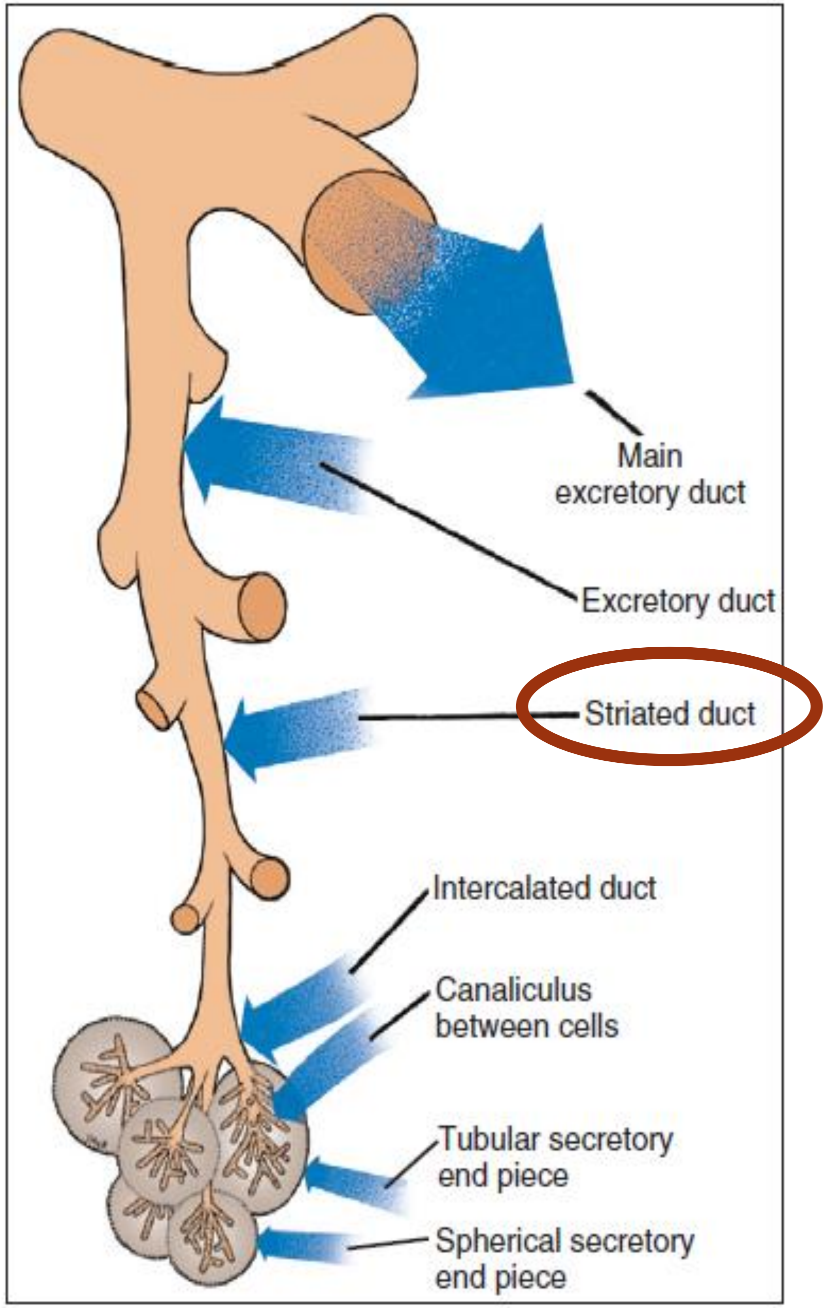
- The smallest (and most distal) of the ducts is the intercalated duct.
- Leads from the serous acini into the striated duct and is usually compressed b/w acini
- contribute to ***primary secretion*** (modify the composition of the saliva (as well as transport it), with moderate activity of respiratory enzymes, as in acini.
- Several acini drain into each intercalated duct.
- In the parotid gland, intercalated ducts are characteristically long, narrow and branching.



DUCTAL SYSTEM

STRIATED DUCTS:

- Intralobular
- Main function: modify the composition of the saliva (as well as transport it)
- form a much longer and more active component of the duct system than the intercalated ducts.
- shows numerous striations under the light microscope.
- They are the site of **electrolyte resorption (especially of sodium and chloride) and secretion (potassium and bicarbonate)** without loss of water.
- As this resorption is against a concentration gradient, it requires substantial amounts of energy.
- The effect on the material in the lumen is to convert an **isotonic or slightly hypertonic fluid (with concentrations similar to those in the plasma) into a hypotonic fluid**.
- There is therefore intense activity of respiratory enzymes in the striated ducts in order to effect and maintain these concentration gradients and the hypotonicity of the fluid in the lumen, as also in the submandibular gland
- The cells of the striated duct exhibit small secretory granules in the luminal region that may contain epidermal growth factor, lysozyme, kallikrein and secretory IgA.
- The granules are less abundant in humans than in many other species an less abundant in the parotid than in the submandibular gland.



DUCTAL SYSTEM

COLLECTING DUCTS:

- The striated duct leads into the collecting duct
- main function of the collecting ducts is to **transport saliva**

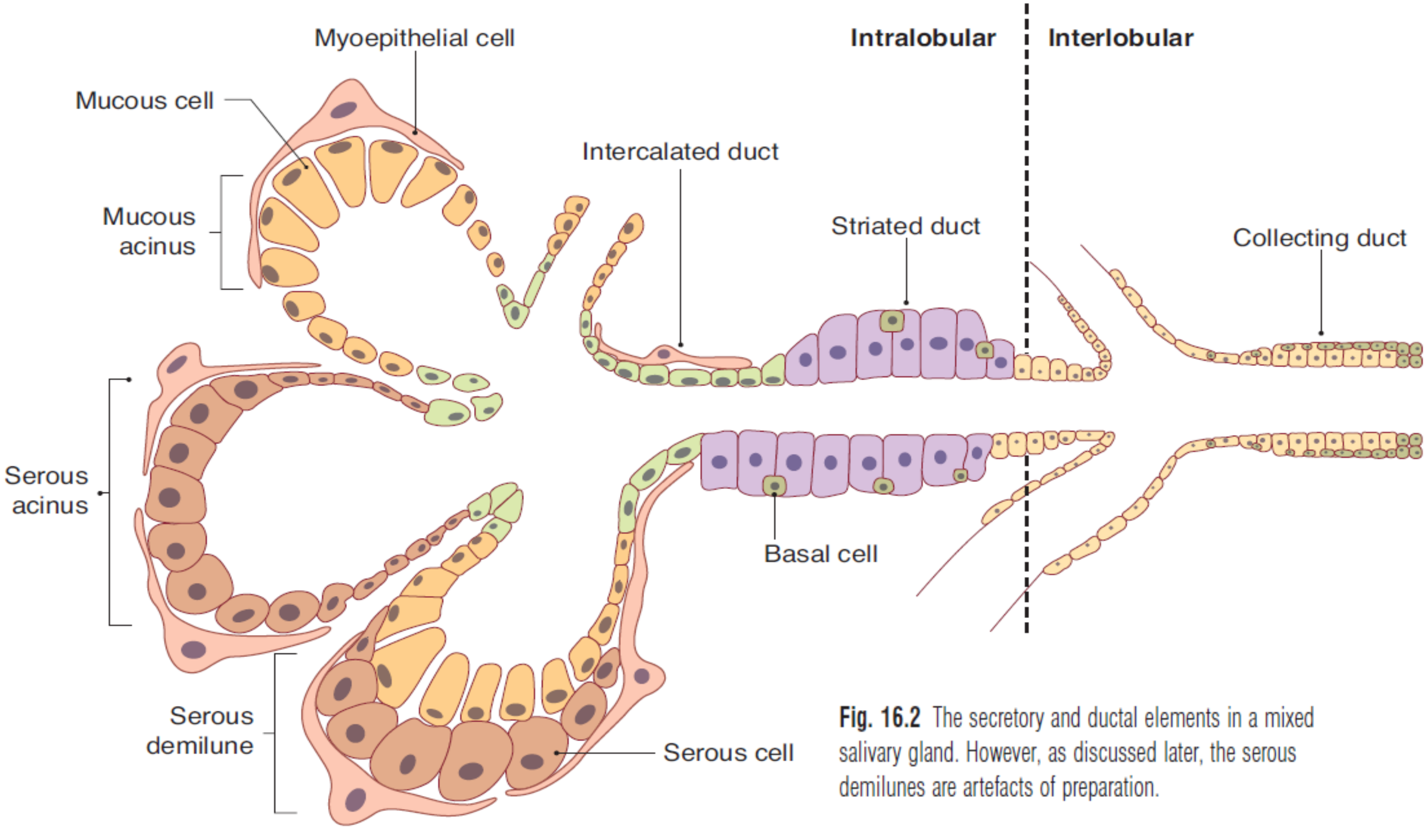
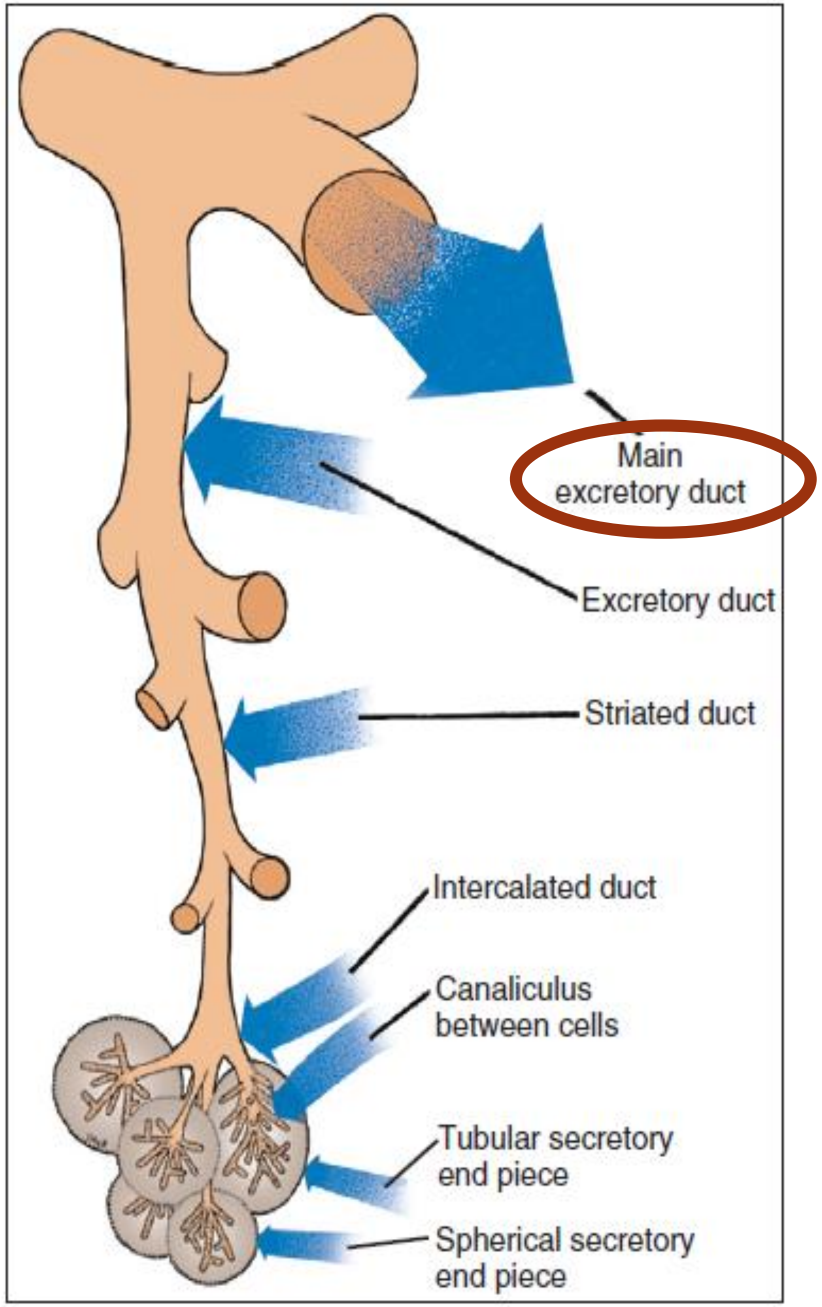
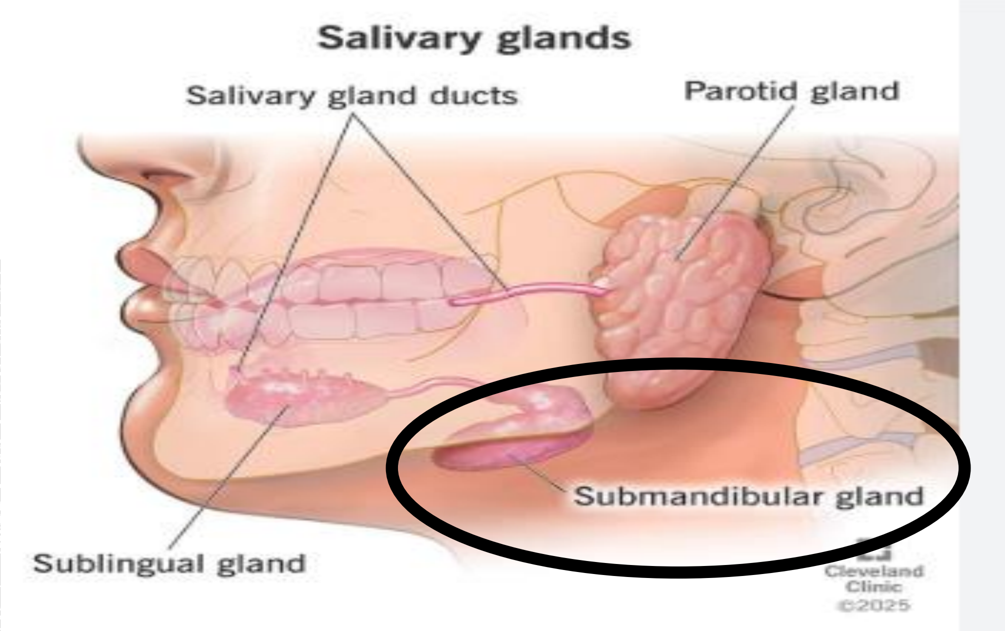


Fig. 16.2 The secretory and ductal elements in a mixed salivary gland. However, as discussed later, the serous demilunes are artefacts of preparation.



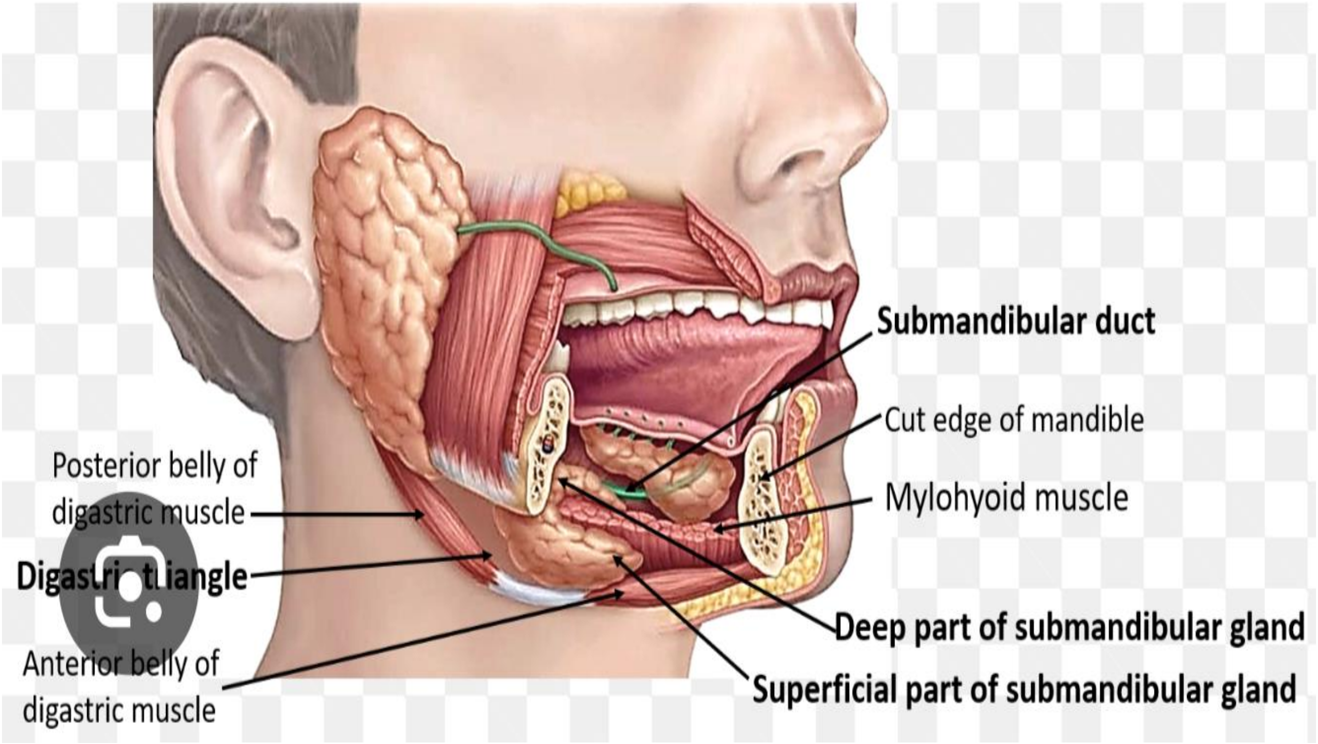
Submandibular Gland

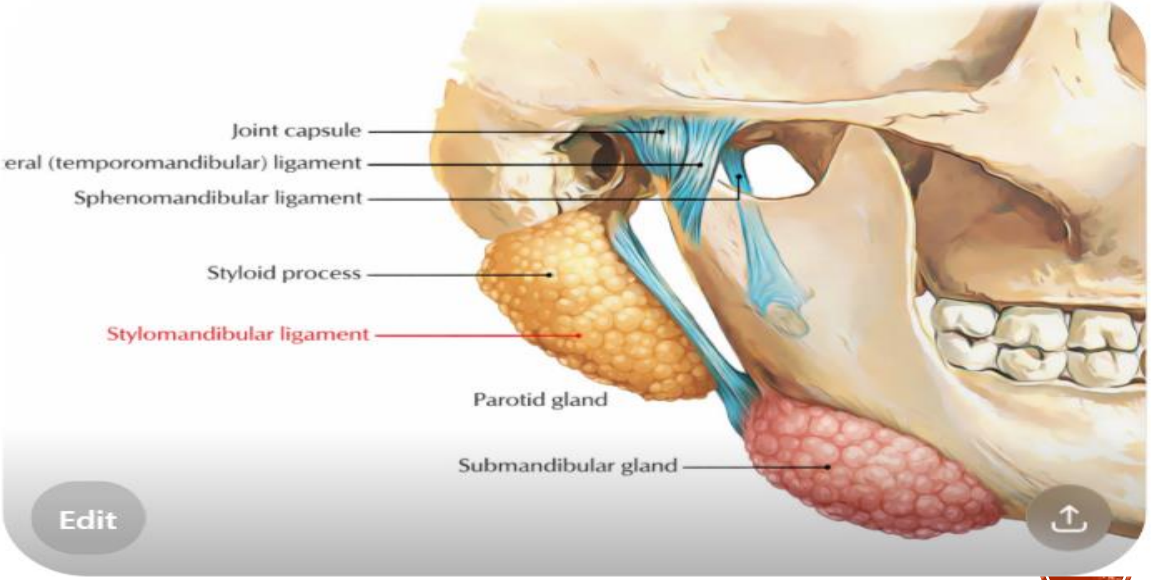
- The second largest of the salivary glands, the submandibular gland,
- produces a mixed serous/ mucous (3:2 ratio) secretion.
- The overall mean of the proportional volume of mucous cells is 8% of the total acinar volume, although it varies between 1% and 33%.
- It has a connective tissue capsule.
- The intercalated ducts are shorter and the striated ducts are longer and more conspicuous.



GROSS ANATOMY

- **Location:** found in the floor of the mouth and in the suprahyoid region of the neck.
- **Superficial Part:** visible just beneath the inferior border of the mandible .
- **Deep Portion:** The gland has an important relationship with the mylohyoid muscle, wrapping around the free posterior border (not unlike the letter C), giving rise to the smaller deep portion of the gland.
- Posteriorly, the submandibular gland comes close to the apex of the parotid gland, with only the stylomandibular ligament intervening.





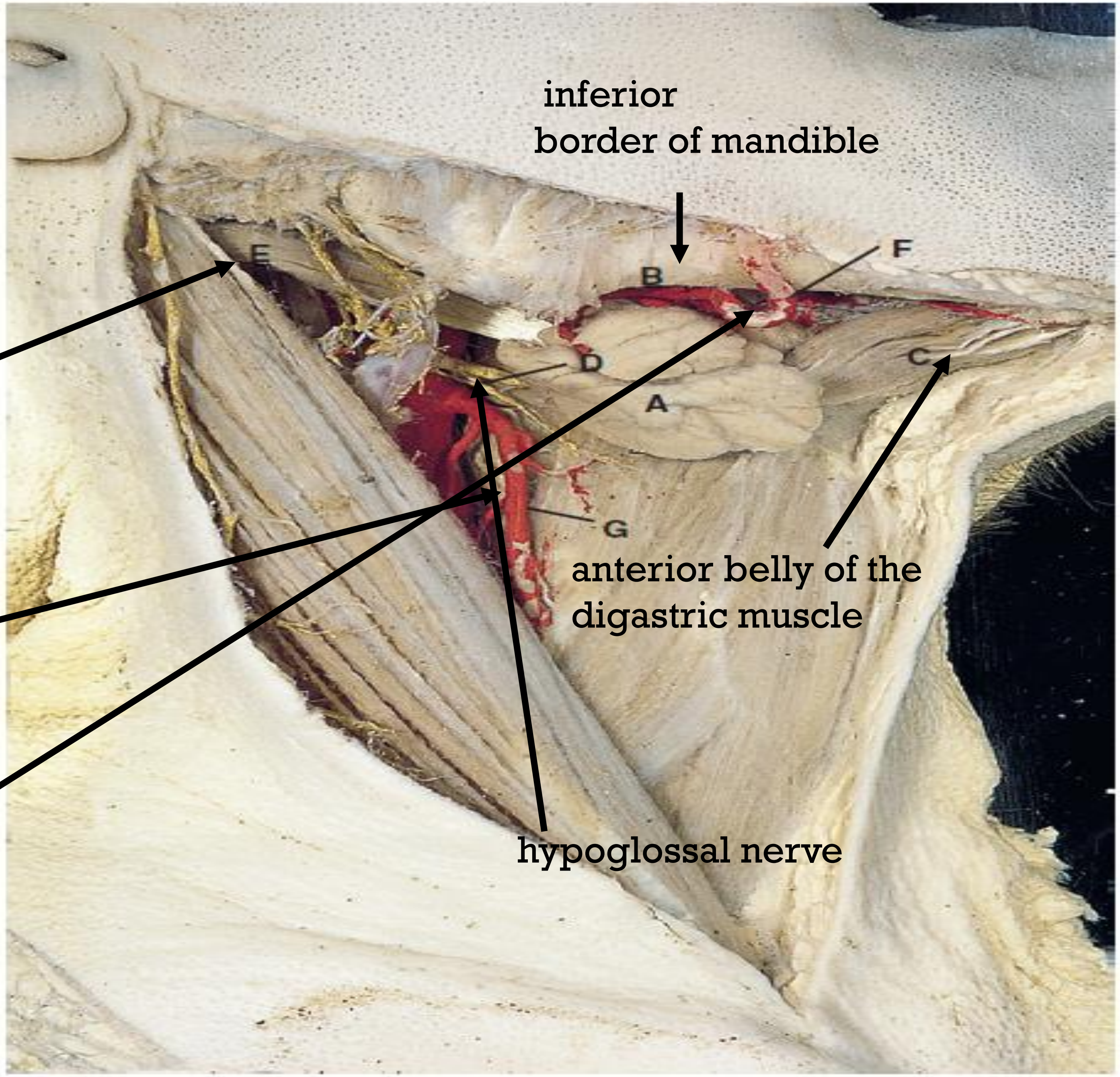
GROSS ANATOMY

- **Location:** found in the floor of the mouth and in the suprahyoid region of the neck.

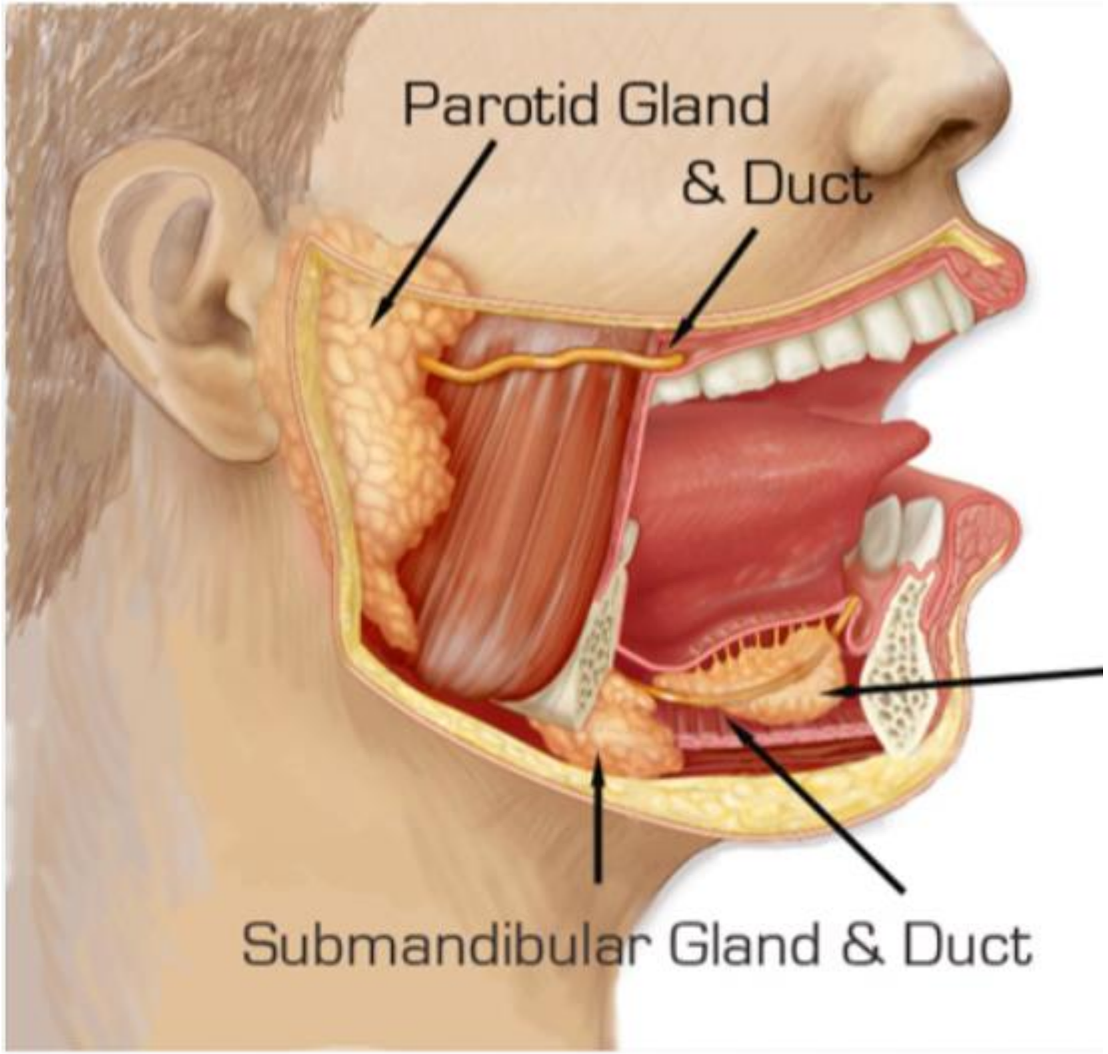
posterior belly of the digastric muscle

superior thyroid artery

facial artery on the mylohyoid muscle



- The submandibular duct (Wharton's duct) appears from the deep part of the gland and wraps around the lingual nerve, as it crosses the hyoglossus muscle, to terminate on the sublingual papilla in the floor of the mouth



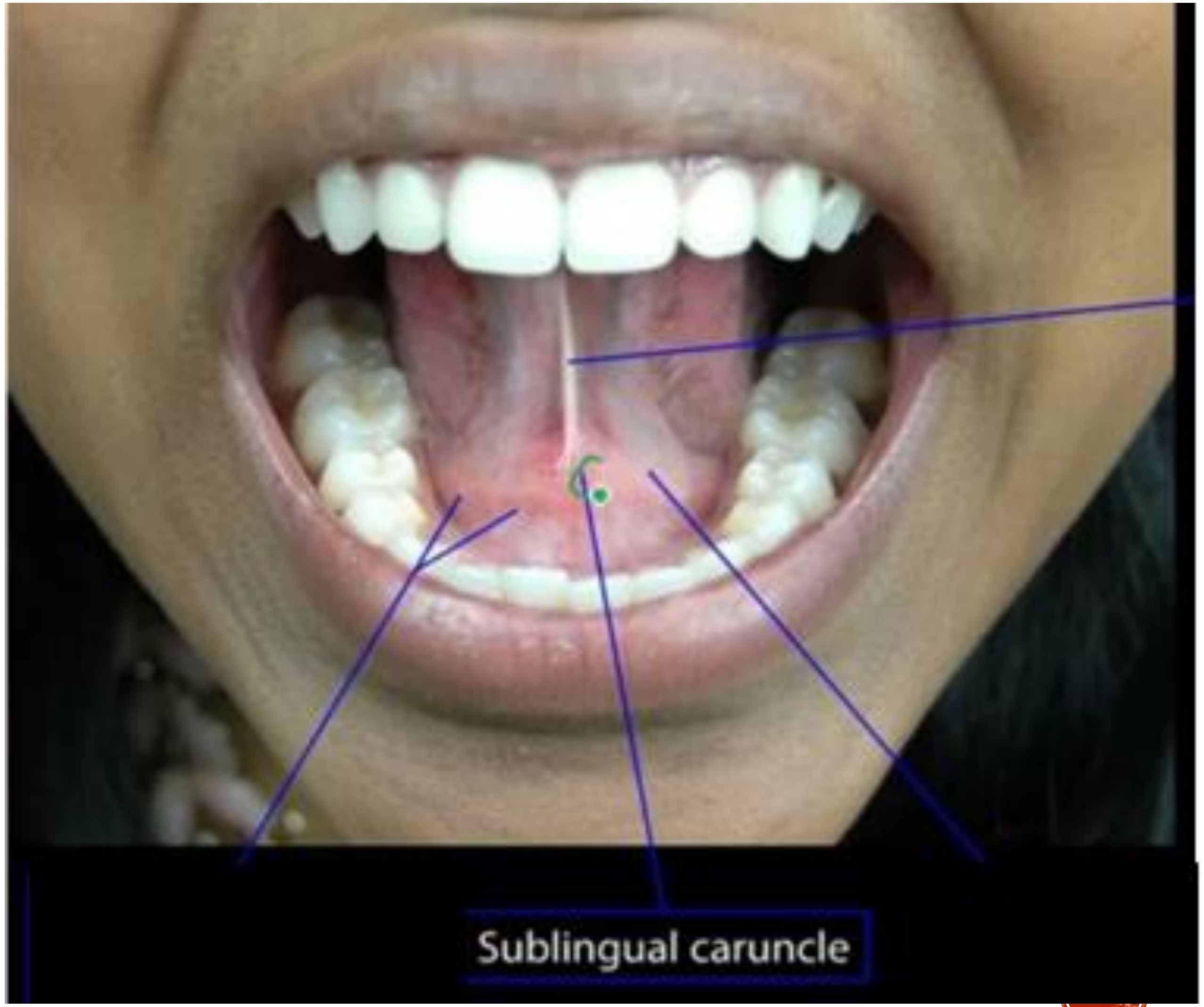


Table 16.1 Comparison of three major salivary glands

Parotid gland	Submandibular gland	Sublingual gland
Largest	Intermediate	Smallest
Serous	Mixed, mainly serous	Mucous
Intercalated ducts	Intercalated ducts	Few intercalated ducts
Striated ducts present	Striated ducts present	No striated ducts
Collecting ducts end in single main duct	Collecting ducts end in single main duct	Collecting ducts end in many main ducts
Nonspontaneous secretor	Nonspontaneous secretor	Spontaneous secretor
Saliva hypotonic	Saliva hypotonic	Saliva isotonic

SUBLINGUAL GLAND

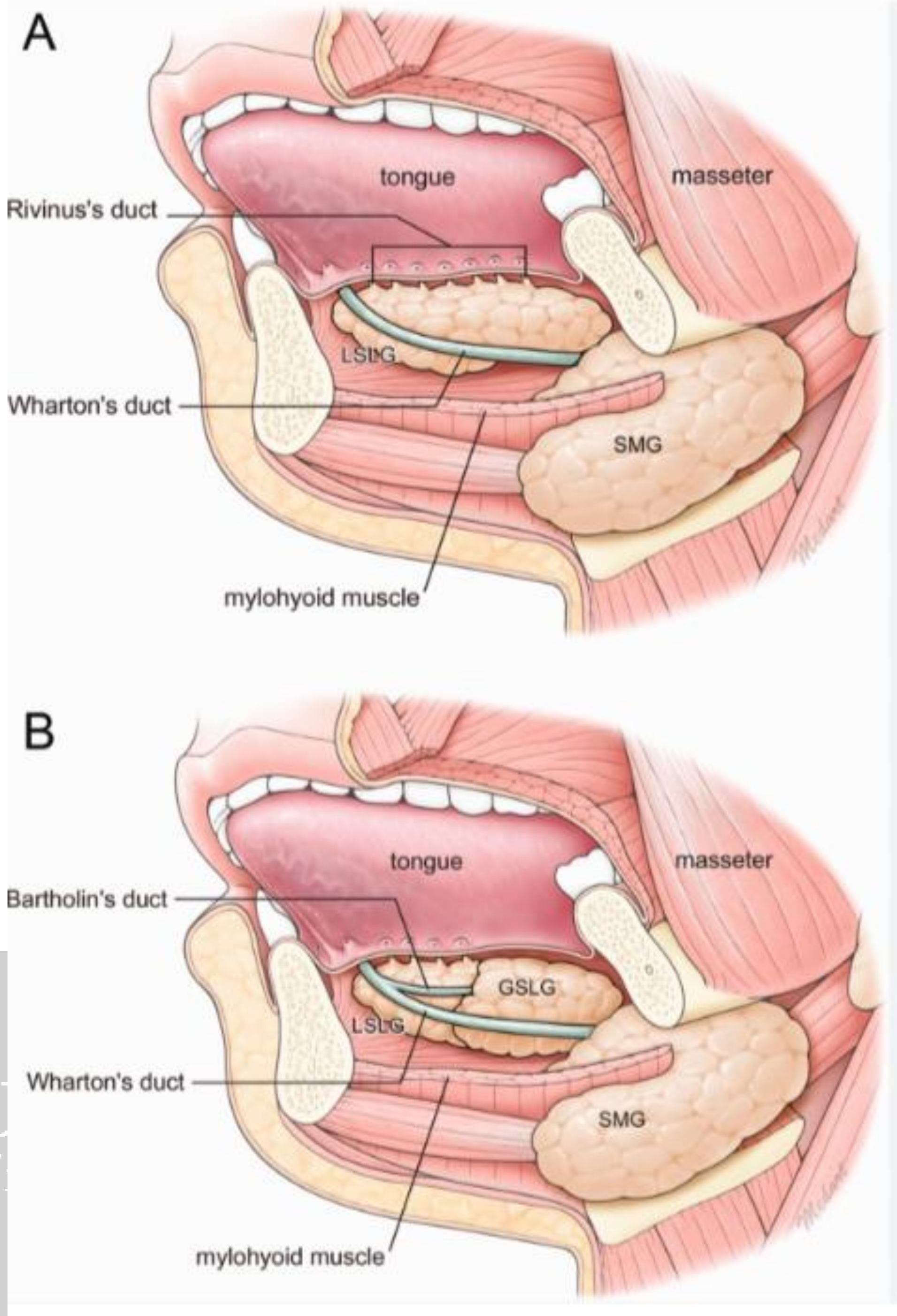
smallest of the three major pairs of salivary glands

not a single unit, unlike parotid & submandibular glands

Posterior part: (the greater sublingual gland) - not always present. Its main duct (Bartholin's duct) leads from it to either join the submandibular duct or to drain directly onto the sublingual papilla.

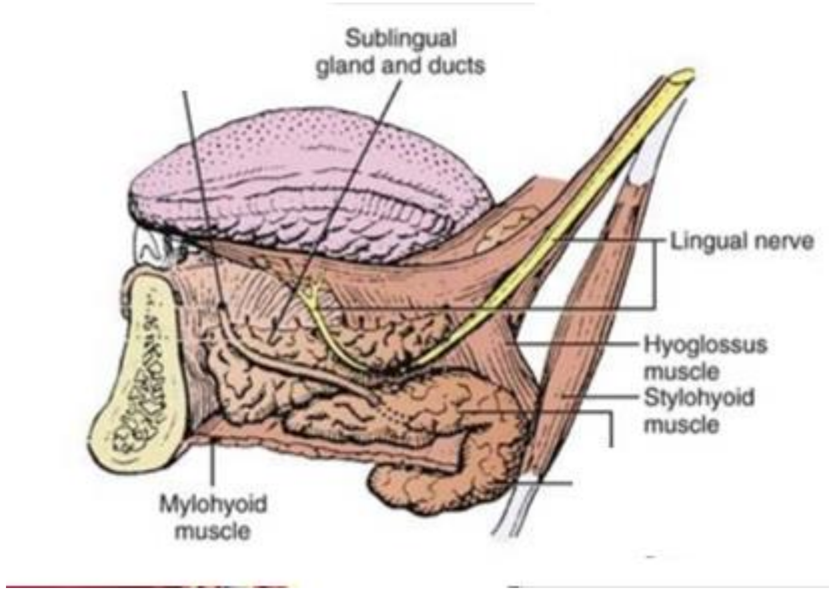
Anterior Part: consists of 7 to 15 small salivary glands, each having its own duct (a Rivinus duct) passes to drain onto the sublingual fold.

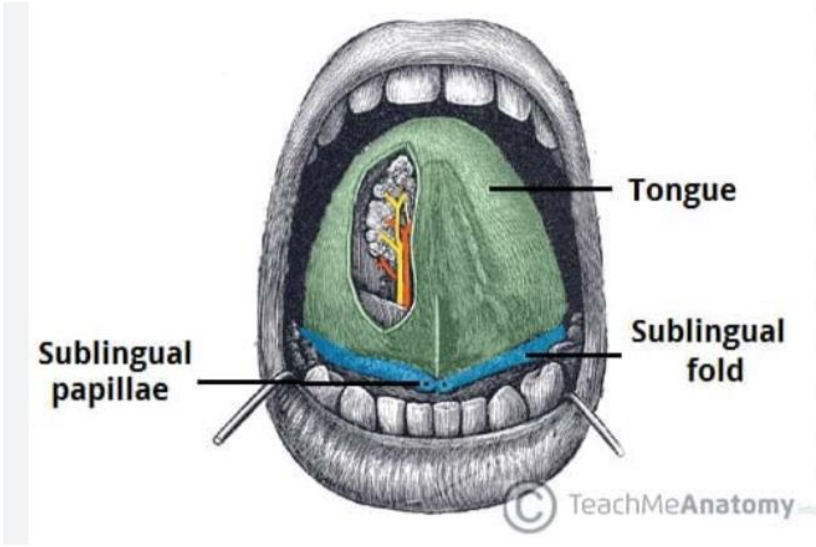
Mixed gland (previous opinions) more recent → mucous



GROSS ANATOMY:

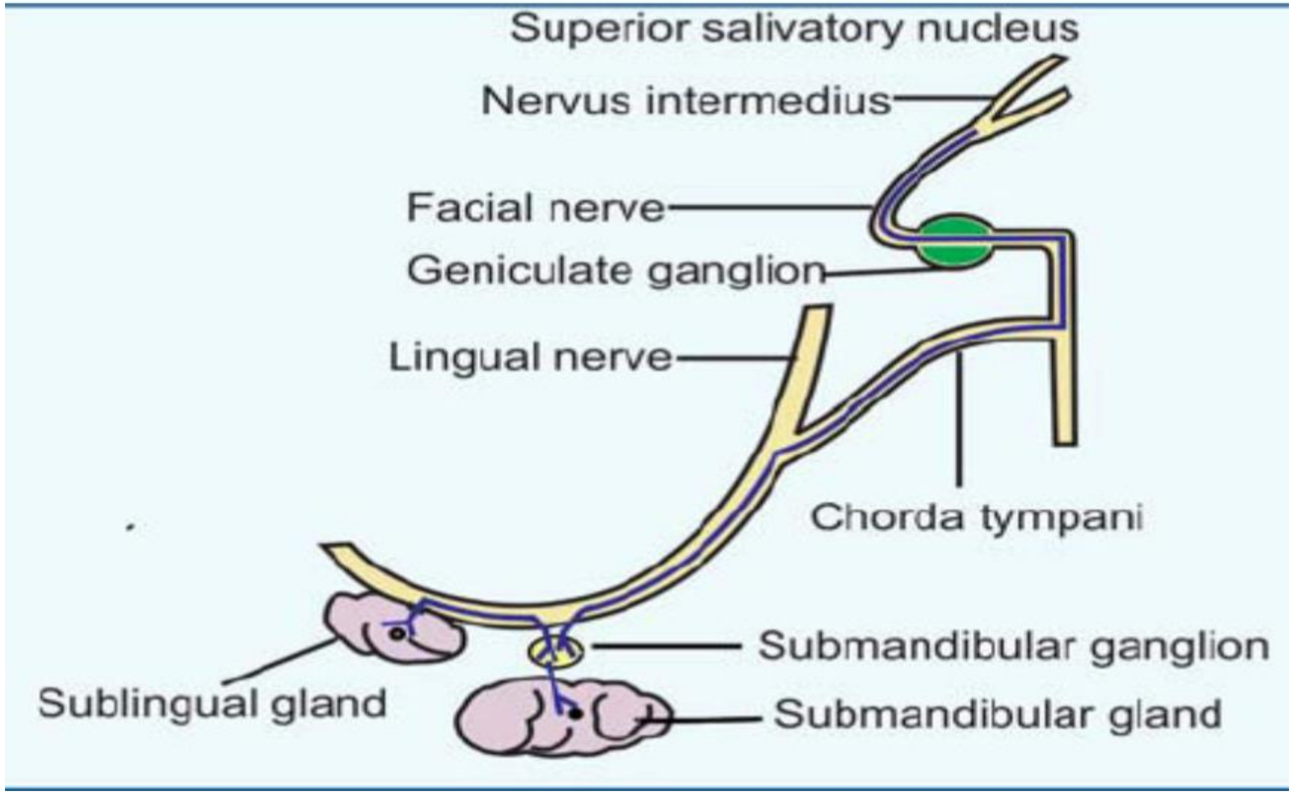
- **Location:** on the hyoglossus muscle in the floor of the mouth, adjacent to the sublingual fossa of the mandible.
- Associated with the sublingual folds beneath the tongue.
- In coronal section, the gland rests on the mylohyoid muscle.
- The posterior part of the sublingual gland may be joined to the deep part of the submandibular gland to form a single sublingual–submandibular complex.





INNERVATION

- The parasympathetic innervation of both the submandibular and sublingual glands is the chorda tympani branch of the facial nerve.
- Preganglionic fibres are carried with this nerve (via the lingual nerve) to the submandibular ganglion.
- Postganglionic fibres pass from this ganglion to the submandibular and sublingual glands



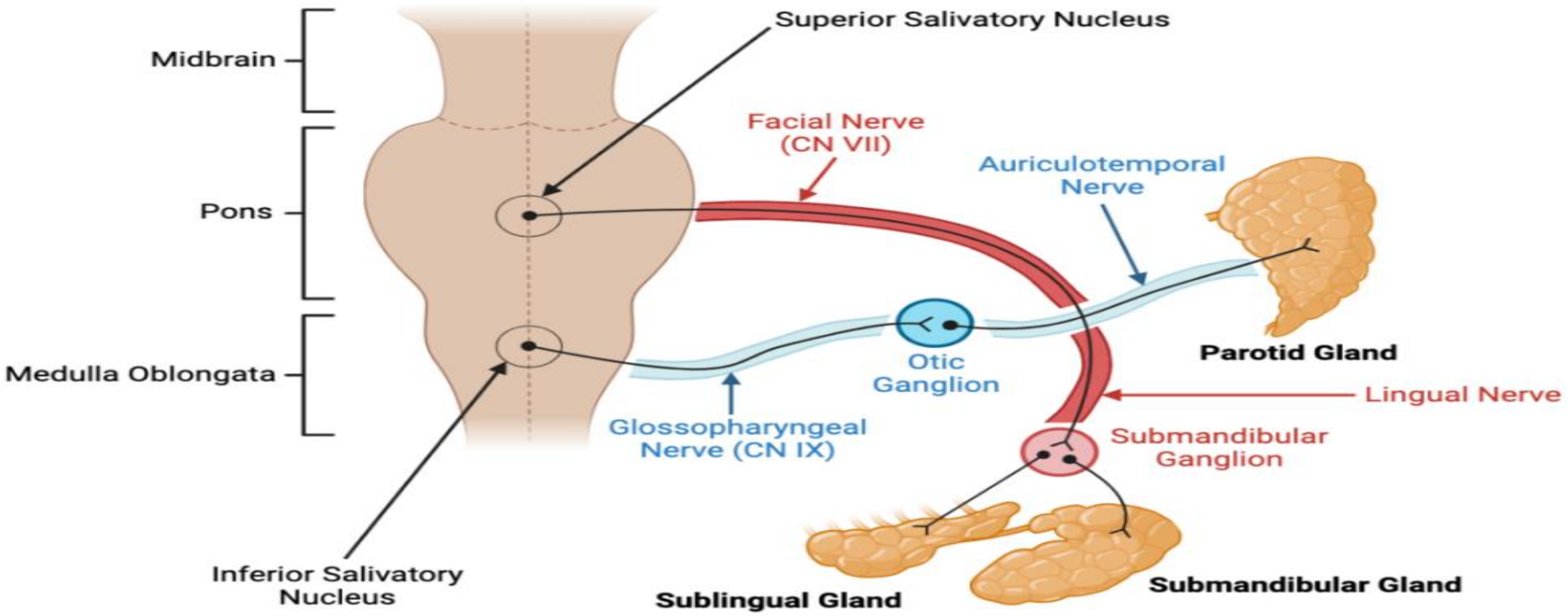


Fig 2

Path of parasympathetic fibres to the salivary glands. Autonomic fibres travel with other cranial nerves such as CN IX, as pictured.

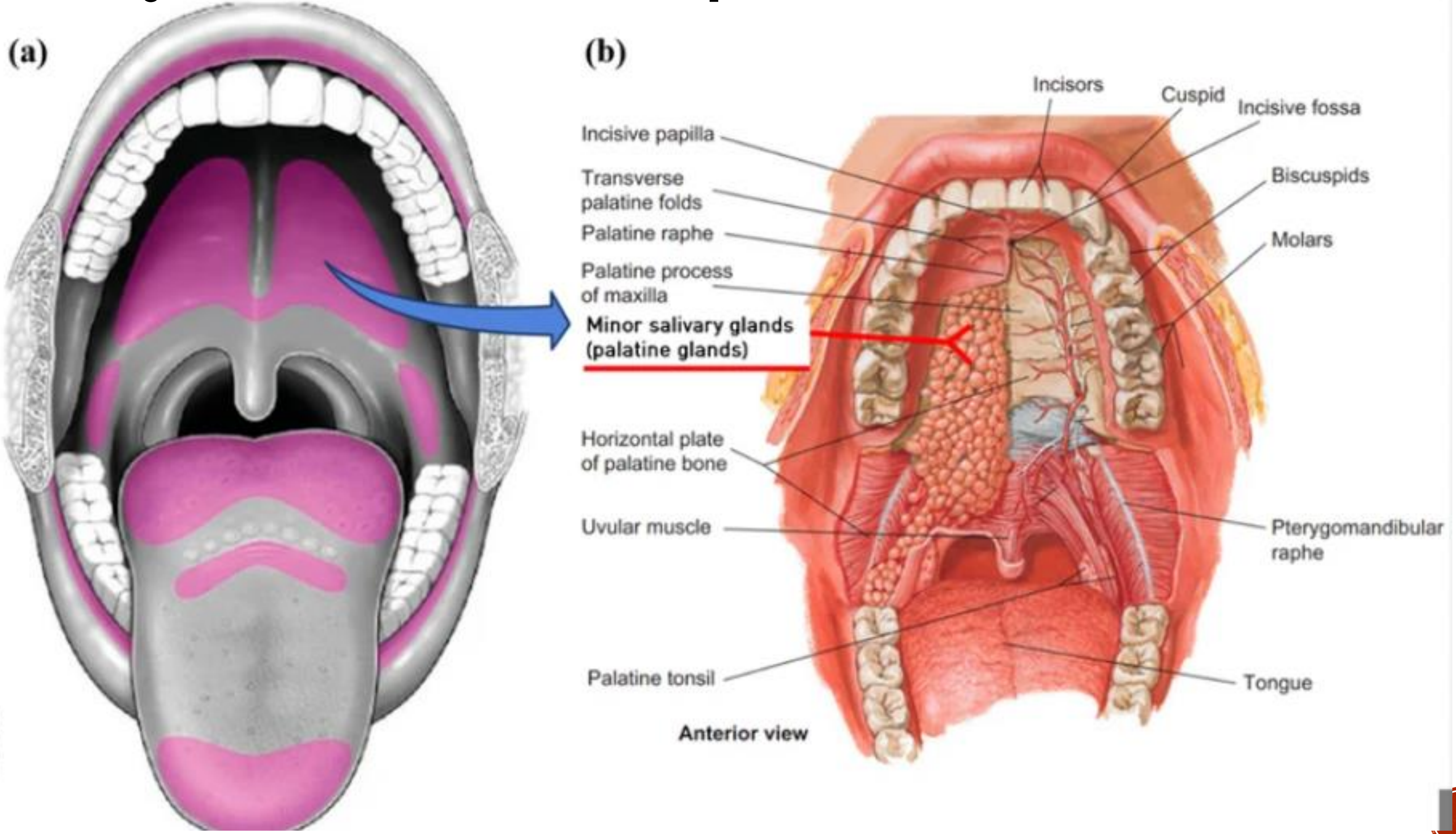
Unlike endocrine glands, the secretion of which is controlled by the activity of hormones, the secretion of saliva by the salivary glands is under the control of the autonomic nervous system.

MINOR SALIVARY GLANDS:

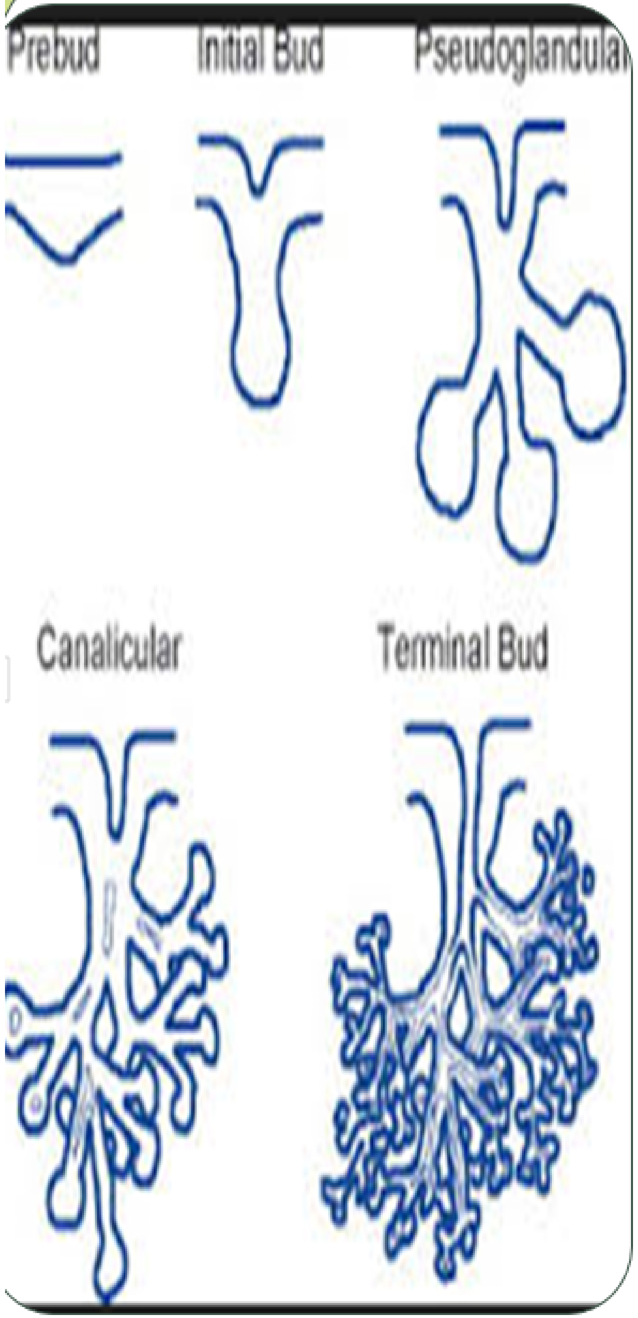
- The minor salivary glands are classified by their anatomical location in the submucosa: buccal, labial, palatal, palatoglossal and lingual.
- b/w 450 and 1,000.
- Pre dominantly- mucous glands except for the serous glands of von Ebner, which drain into the trench of the circumvallate papillae.
- The anterior lingual glands are embedded within muscle near the ventral surface of the tongue and have short ducts opening near the lingual frenum.
- The posterior glands are located in the root of the tongue.
- Minor salivary glands have collecting ducts, but intercalated and striated ducts are usually absent.
- This duct system therefore does not usually remove salt, so the final saliva released from them is isotonic and rich in sodium.
- sympathetic nerves sparsely innervate minor salivary glands, and so predominant parasympathetic nerve impulses.
- However, the minor salivary glands secrete spontaneously and continuously, as do the sublingual glands.

Not present in gingiva & ant. 2/3rds of tongue.

The palatoglossal glands are located in the region of the pharyngeal isthmus. glands lie in both the soft and the hard palate.



SALIVARY GLAND DEVELOPMENT



Development of Salivary Glands

- **Development Timeline →** 6th & 7th weeks IU

Develops from reciprocal epithelium – mesenchymal interactions

- **Origin →**
  Solid buds : Oral epithelial-derived Solid buds extending from the developing mouth into the underlying mesenchyme
  Surrounding Connective Tissue: neural crest
- **Formative Elements →** ducts & secretory elements of salivary glands


Salivary Glands- histology
Info
Watch later
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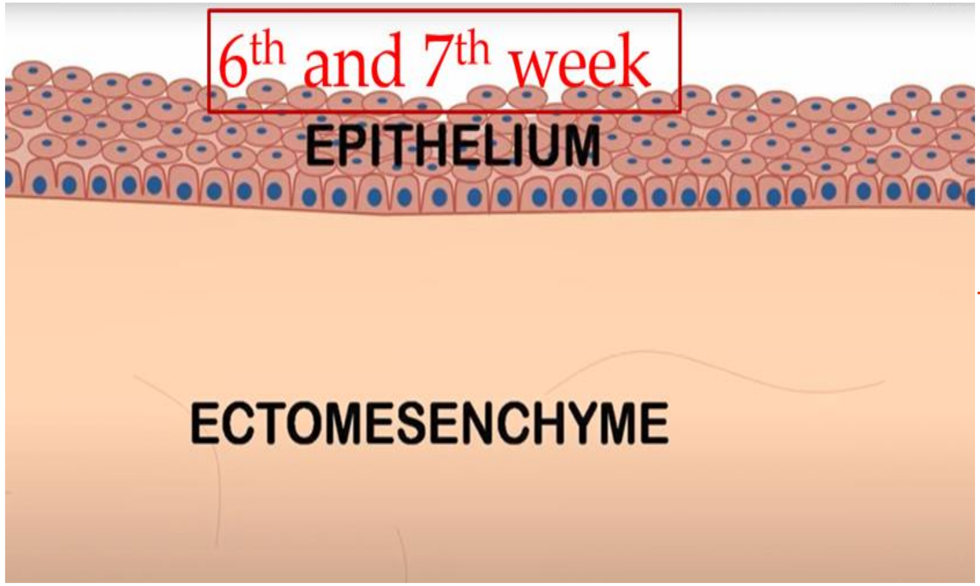
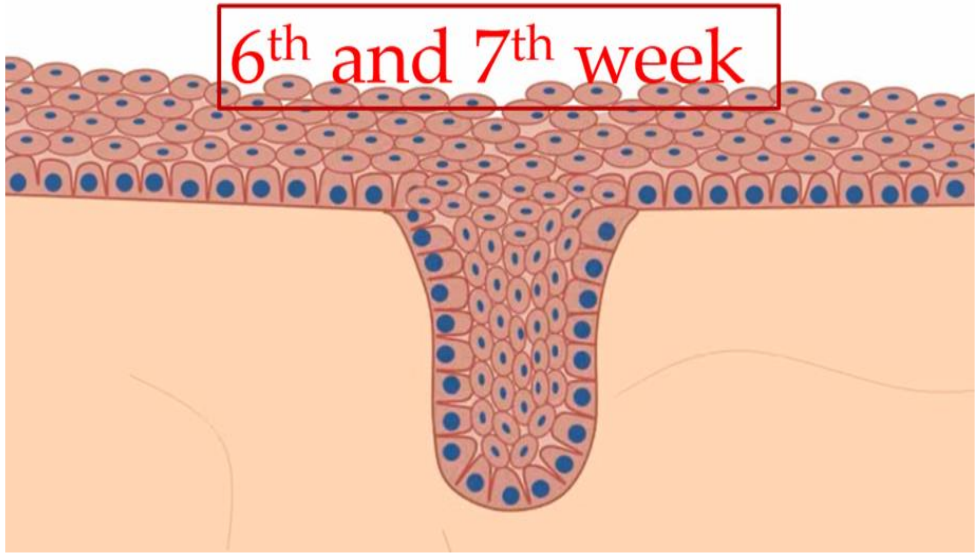
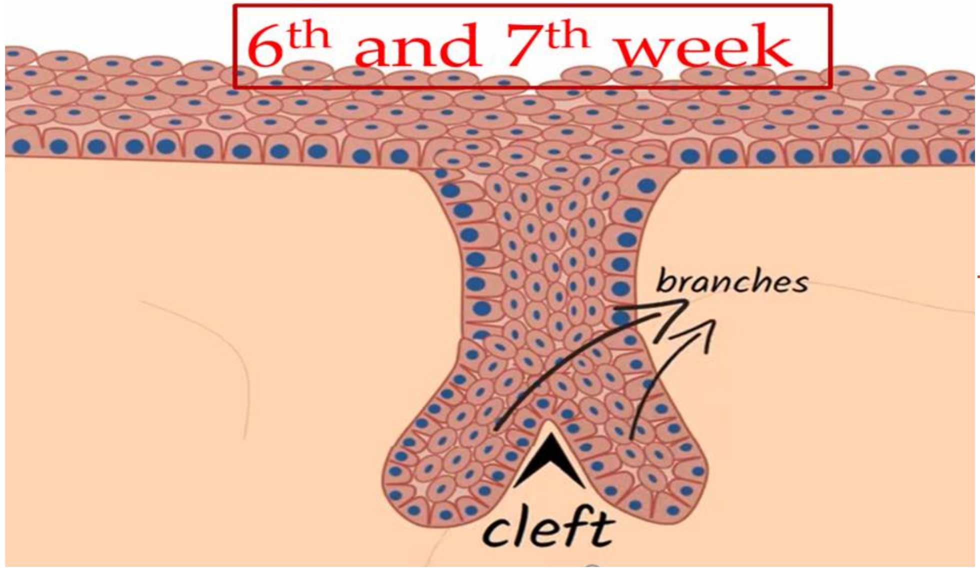
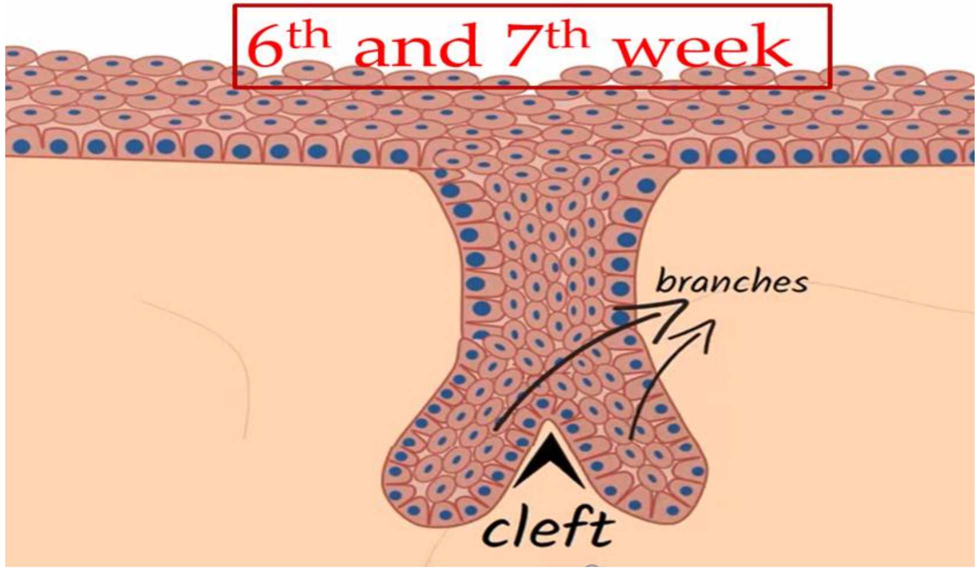
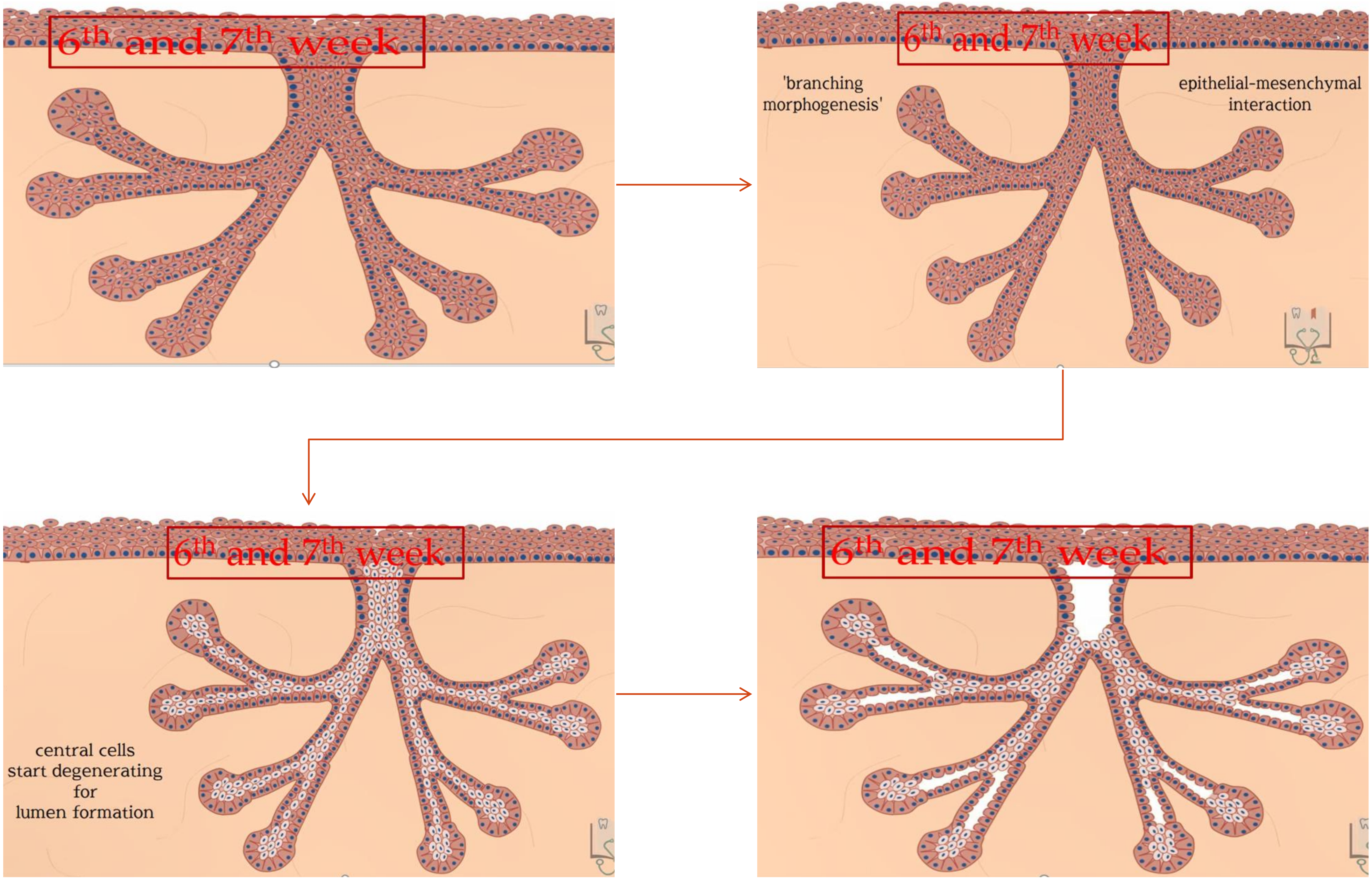
DEVELOPMENT
MORE VIDEOS
0:25 / 9:14
YouTube

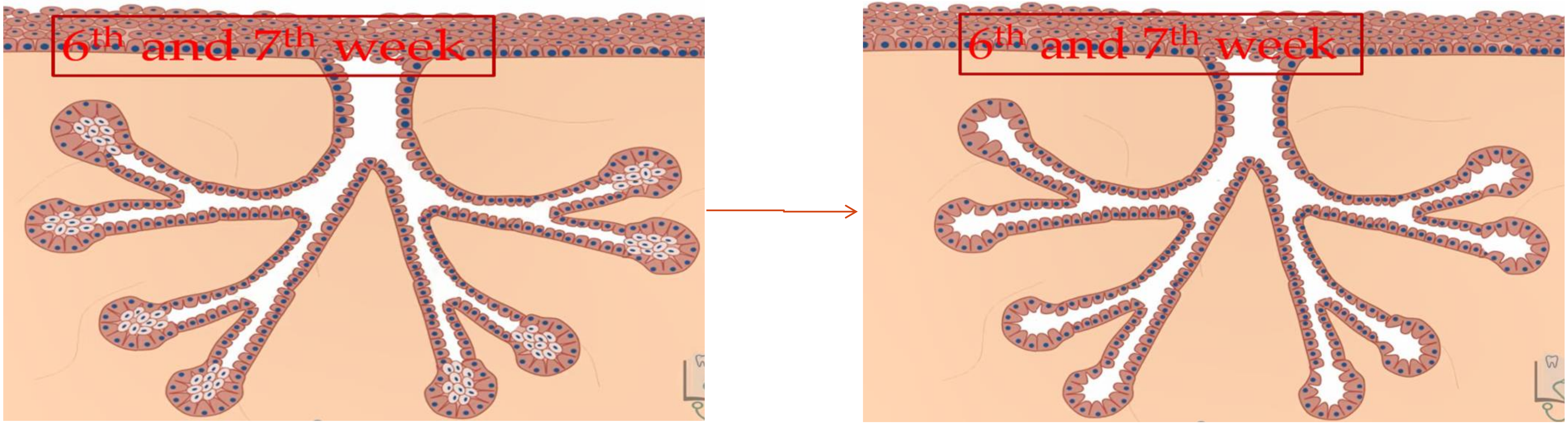

6th and 7th week
EPITHELIUM
ECTOMESENCHYME


6th and 7th week


6th and 7th week
branches
cleft


6th and 7th week
branches
cleft


6th and 7th week
6th and 7th week
'branching morphogenesis'
epithelial-mesenchymal interaction
6th and 7th week
central cells start degenerating for lumen formation
6th and 7th week



GENES, GROWTH FACTORS AND THEIR RECEPTORS FOR NORMAL SALIVARY GLAND DEVELOPMENT:

FGF

Shh

Pitx2

platelet-derived growth factor (PDGF)

TGFß

EGF

BRANCHING MORPHOGENESIS"

- Clefts develop in the bud, forming two or more new buds; continuation of this process, called branching morphogenesis, produces successive generations of buds and a hierarchic ramification of the gland.

SEQUENCE OF LUMEN DEVELOPMENT

The development of a lumen within the branched epithelium generally occurs in this order:

(1) in the distal end of the main cord and in branch cords,

(2) in the proximal end of the main cord, and

(3) in the central portion of the main cord.

- The lumina form within the ducts before they develop within the terminal buds.

FORMATION OF MUCOUS, SEROUS AND MYOEPITHELIAL CELLS

After lumen development (terminal buds) → epithelium is 2 layered

Inner layer cells → differentiate into the secretory cells of the mature gland:

- mucous
- serous

Outer layer cells → contractile myoepithelial cells (present around the secretory end pieces and intercalated ducts)

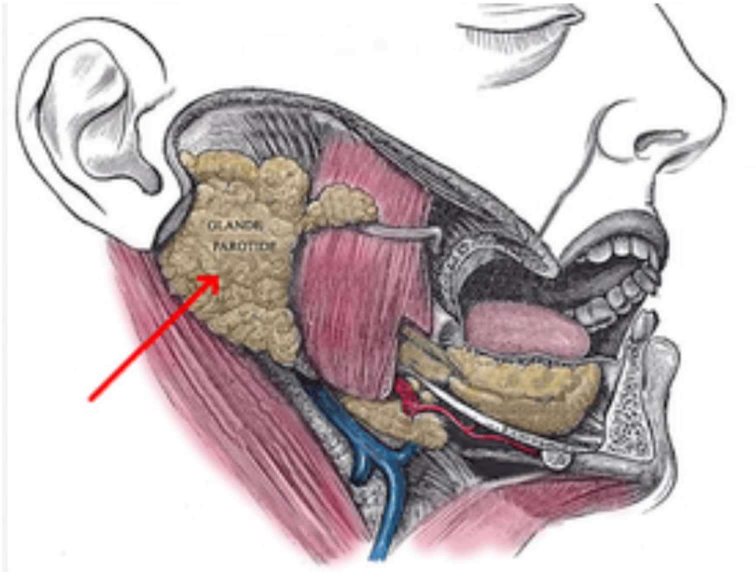
FORMATION OF MUCOUS, SEROUS AND MYOEPITHELIAL CELLS

Epithelial parenchymal components ↑ size and number → ↓ associated mesenchyme (connective tissue) → remaining thin layer of CT surrounds each secretory end piece and ducts.

Thicker partitions of CT (septa), continuous with the capsule (within which run the nerves and blood vessels that

supplying the gland). These function to

- invest the excretory ducts
- divide the gland into lobes and lobules

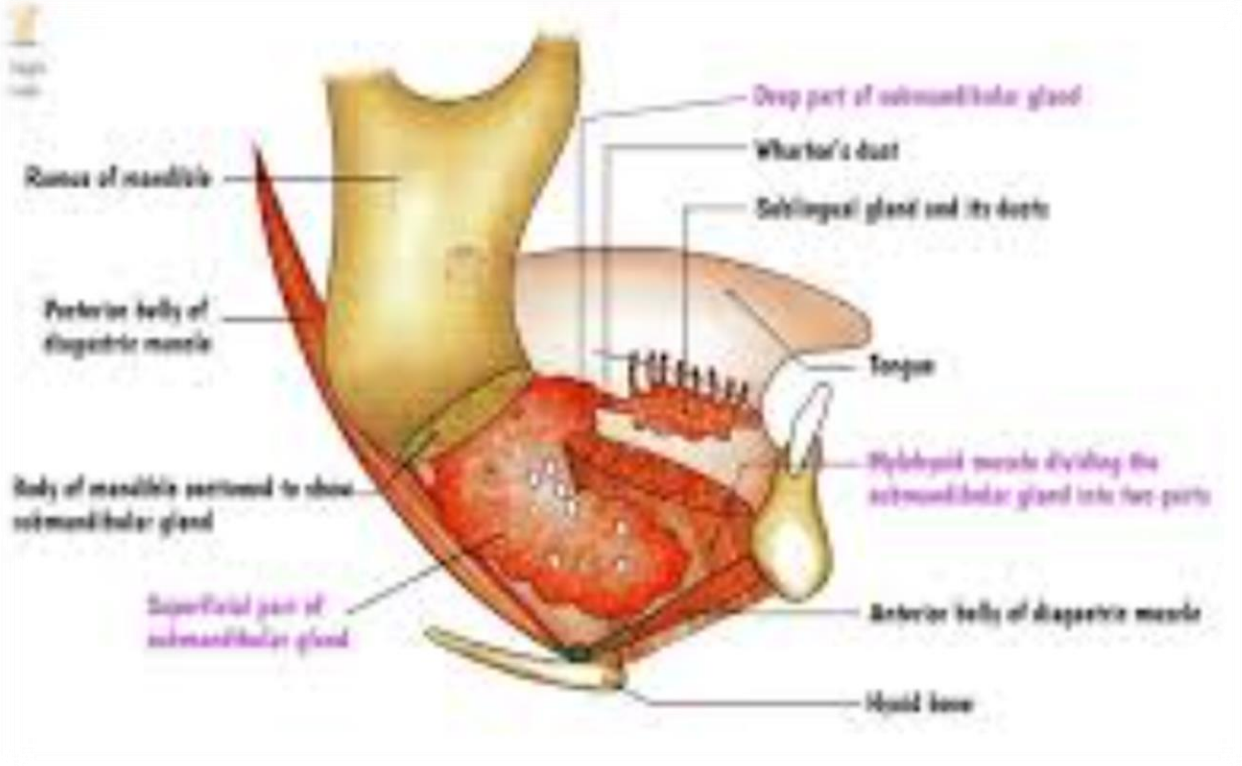


1ST TO DEVELOP

PAROTID GLAND TIMELINE

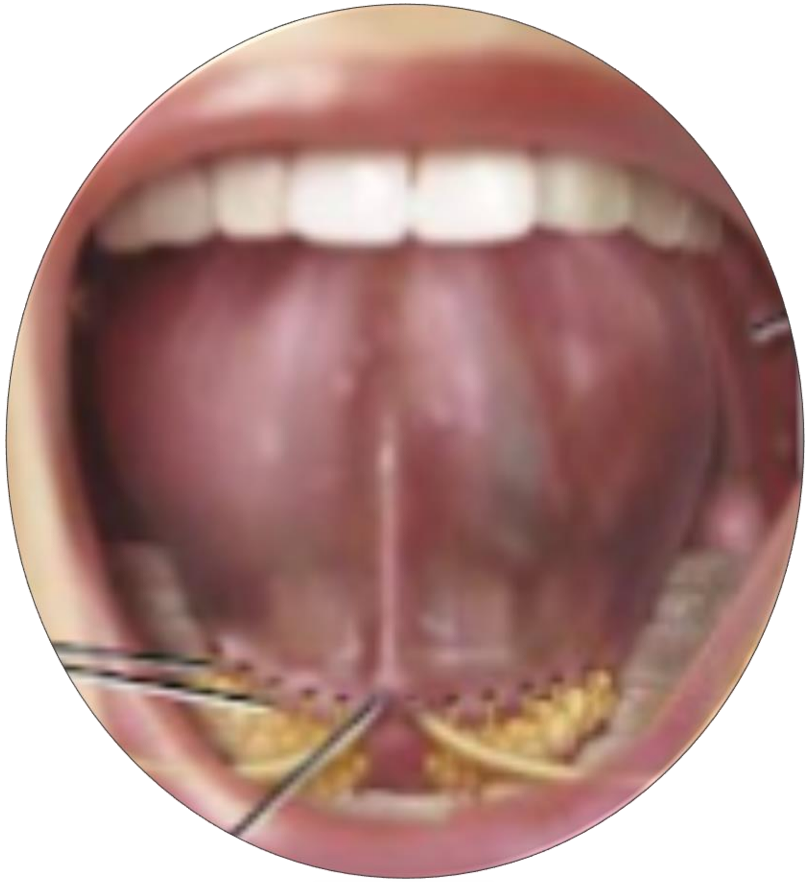
- Near angles of mouth (stomodeum).
- Epithelial bud for it, extends towards developing ears & branches to form solid cords.
- Canalization → 10th week
- Cord rounded ends → secretory acini (10th week)
- Functional 18th week.

SUBMANDIBULAR GLAND TIMELINE



2^{ND} TO DEVELOP

- Later than parotid, in 6^{TH} week IU.
- Epithelial bud growth → post. & lateral to tongue
- Canalise → 12^{th} week (formation of ducts & acini)
- Functional → 16^{th} week IU



SUBLINGUAL GLAND TIMELINE

- Appear → 8^{th} week
- Multiple epithelial buds -→ paralingual sulcus of forming mouth
- Canalise to form 12 ducts to open independently into floor of mouth

Thank you!